SCHOLAU

Statistics
CBT & Gap Practice

300 MCQs • 100 fill-in-the-gap questions

STA 111 • STA 112 • STA 121

Prepared from the supplied textbook photographs and handwritten supplement. Structured for topic-by-topic study, with checked worked examples and explicit source issues.

Study edition | 28 October 2026

Original practice material for Adeleke University students. It is not an official university publication, an authenticated past-question collection or a prediction of examination content.

How to use this edition

Source labels and page references

T means STATISTICS TEXTBOOK CH 1 - 8.pdf (133 PDF pages). S means STA 112.pdf (70 PDF pages). T18 means the 18th page in the PDF, counting from 1; T18-21 means pages 18 through 21. Textbook printed page numbers generally match PDF positions. Supplement references use PDF positions because handwritten page labels are not consistent.

Course allocation

Course	Verified source coverage
STA 111	1 • Concepts of statistics | T1-9
STA 111	2 • Descriptive statistics | T10-47
STA 112	3 • Random variables | T48-55
STA 112	4 • Probability theory | T56-72
STA 112	5 • Probability distribution | T73-94
STA 112	Supplement • all relevant statistics material | S1-66, S69-70
STA 121	6 • Sample techniques | T95-103
STA 121	7 • Hypothesis testing | T104-120
STA 121	8 • Confidence interval and estimation theory | T121-133

Coverage and editorial policy

All 203 page images were reviewed. The pack reorganises and explains the relevant statistics content; it is not a line-by-line facsimile. Duplicated examples are consolidated. Original examples and pedagogical clarifications are identified by context. Definite mathematical errors are corrected; uncertain text and missing data are flagged rather than invented. The lesson identifiers provide a stable course > topic > lesson structure for ScholAu study organisation; no unprovided upload schema or platform integration is claimed.

Equations and conventions

Use the stated model and units. Unless a question specifies otherwise, retain full precision until the final answer. Numeric MCQs state their rounding. Quantiles use explicitly stated rules. Normal notation $N(\mu,\sigma^2)$ gives variance as its second parameter. Gamma uses shape and rate. A given critical value in a question is part of its inputs.

Source status

No author or textbook title page was supplied. The filenames identify the local sources. The supplement visibly mixes course labels and includes unrelated pages; the detailed map and issue log in the lesson PDF explain the treatment. Source-document assignments and incidental instructions are treated as content to study, not as directions overriding this request.

Practice instructions

This bank contains 300 four-option MCQs and 100 fill-in-the-gap items. Choose one best answer for each MCQ. Answers and explanations are in a separate part after all questions. The compact key is provided for marking, followed by worked solutions with source references. Sources support the tested methods; original numbers are not claimed to be quoted from the scans. Page references beside questions let you return to the relevant source.

Suggested marking: one mark per correct MCQ; one per gap, accepting mathematically equivalent forms and standard alternative terminology. Use a calculator where needed. This is a practice bank, so there is no imposed time limit or negative marking.

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Part I • Questions

STA 111

Topic 01 • Statistical thinking and inference

Source pathway: T1-4

Q001 Reporting the observed mean score of a class is primarily which activity?

A. Randomisation
B. Descriptive statistics
C. Population enumeration
D. Inductive inference

Source: T1-4

Q002 Using a random sample to estimate a population mean is which reasoning?

A. Inductive
B. Deductive only
C. Circular
D. Purely descriptive

Source: T1-4

Q003 Which quantity describes a population rather than a sample?

A. Sample statistic
B. Sample mean only
C. Observed relative frequency only
D. Parameter

Source: T1-4

Q004 Which is the most defensible first step in an investigation?

A. Calculate a mean immediately
B. Discard unusual values
C. Define the question and target population
D. Choose the largest answer

Source: T1-4

Fill in the gaps

F001 A numerical characteristic of a population is a _____.

Source: T1-4

F002 Reasoning from sample evidence to a population is _____ inference.

Source: T1-4

Topic 02 • Variables and the meaning of data

Source pathway: T4-6, T9

Q005 A hospital identification number is best treated as what?

A. A continuous measurement
B. A categorical identifier
C. A population mean
D. A quantitative count of illness

Source: T4-6, T9

Q006 Which is naturally modelled as continuous?

A. Exact elapsed time
B. Number of goals
C. Number of clinics
D. Number of siblings

Source: T4-6, T9

Q007 Which statement defines a discrete variable most generally?

A. It must have only positive integers
B. It must be categorical
C. It has no probability distribution
D. It has countably many possible values

Source: T4-6, T9

Q008 Age recorded at last birthday is what type of measurement?

A. Nominal identifier
B. Continuous as recorded
C. Discrete quantitative
D. Purely qualitative

Source: T4-6, T9

Fill in the gaps

F003 A number used only to label a hospital record is an _____.

Source: T4-6, T9

F004 A variable with countably many possible values is _____.

Source: T4-6, T9

Topic 03 • Data sources, collection and quality

Source pathway: T5-8

Q009 An analyst reuses an existing published census table. To that analyst it is what?

A. A sample-space outcome
B. Secondary data
C. A new random assignment
D. Primary experimental data

Source: T5-8

Q010 Which collection approach deliberately manipulates a factor?

A. An experiment
B. A census by definition
C. Passive observation
D. Documentary inspection only

Source: T5-8

Q011 Which issue can produce nonresponse bias?

A. A variable has units
B. All selected people respond
C. All units have a valid record
D. Response depends on the characteristic being studied

Source: T5-8

Q012 Which method is most directly suited to exploring group attitudes in depth?

A. A random-number table
B. A standard deviation
C. A moderated focus group
D. A frequency formula

Source: T5-8

Fill in the gaps

F005 Data collected by an investigator for the current purpose are _____ data.

Source: T5-8

F006 A study that deliberately changes a factor is an _____.

Source: T5-8

Topic 04 • Official statistics and responsible use

Source pathway: T8-9

Q013 Which source example is associated with Nigerian financial statistics?

A. A histogram axis
B. Central Bank of Nigeria
C. A class median
D. A sample quartile

Source: T8-9

Q014 Which item is essential when reporting a percentage?

A. Its denominator or reference group
B. Only the largest count
C. The sum of unrelated means
D. Only its colour

Source: T8-9

Q015 Why should a table state its reference period?

A. To guarantee causality
B. To remove all measurement error
C. To force equal group sizes
D. To make the time coverage clear

Source: T8-9

Q016 Who can use official statistics according to the source?

A. Only statisticians
B. Only the original collector
C. Government, researchers, businesses and the public
D. Only banks

Source: T8-9

Fill in the gaps

F007 A reported percentage should identify its reference group or _____.

Source: T8-9

F008 The source names the Central Bank of Nigeria by the abbreviation _____.

Source: T8-9

Topic 05 • Frequency tables and statistical graphs

Source pathway: T10-15, T43-47

Q017 Relative frequency for a class with count f out of N is what?

A. N/f
B. f/N
C. f+N
D. f-N

Source: T10-15, T43-47

Q018 A less-than ogive plots cumulative counts against which class values?

A. Upper class boundaries
B. Class labels in random order
C. Only lower class limits
D. Class frequencies alone

Source: T10-15, T43-47

Q019 For unequal class widths, histogram area represents frequency when height equals what?

A. Class midpoint
B. Cumulative percentage only
C. Frequency times class width
D. Frequency divided by class width

Source: T10-15, T43-47

Q020 Which graph best displays frequencies of named categories?

A. An unlabelled ogive
B. A t critical region
C. Bar chart
D. Continuous density curve by definition

Source: T10-15, T43-47

Q021 A class contains 11 observations out of 45. What is its relative frequency in percent? Give the value to 4 decimal places.

A. 12.2222
B. 24.4444
C. 75.5556
D. 48.8889

Source: T12-13

Q022 A class contains 8 observations out of 40. What is its relative frequency in percent? Give the value to 4 decimal places.

A. 20.0000
B. 10.0000
C. 80.0000
D. 40.0000

Source: T12-13

Q023 A class contains 18 observations out of 60. What is its relative frequency in percent? Give the value to 4 decimal places.

A. 60.0000
B. 70.0000
C. 15.0000
D. 30.0000

Source: T12-13

Q024 T12 frequencies are 3,4,8,11,9,6,4. What is cumulative frequency through the fourth class?

A. 11
B. 35
C. 26
D. 15

Source: T10-15, T43-47

Q025 For observations rounded to whole units, what is the midpoint of class 30-39? Give the value to 4 decimal places.

A. 9.0000
B. 34.5000
C. 39.0000
D. 30.0000

Source: T12-13

Fill in the gaps

F009 Relative frequency is class frequency divided by the _____ frequency.

Source: T10-15, T43-47

F010 A less-than cumulative frequency curve is called an _____.

Source: T10-15, T43-47

F011 Frequencies 3,4,8,11 have cumulative count _____.

Source: T10-15, T43-47

Topic 06 • Stem-and-leaf displays and box plots

Source pathway: T14-17, T44

Q026 The width of the box in a horizontal box plot represents what?

A. Interquartile range
B. Population probability
C. Mean absolute deviation
D. Full sample size

Source: T14-17, T44

Q027 Which quantity determines the usual inner outlier fences?

A. Three times the median
B. 1.5 times the mean
C. The sample count alone
D. 1.5 times the IQR

Source: T14-17, T44

Q028 In a modified box plot, a whisker usually ends at what?

A. Zero in every dataset
B. The fence even if no point lies there
C. The most extreme observation within the fence
D. The arithmetic mean

Source: T14-17, T44

Q029 What must accompany a stem-and-leaf display?

A. A fixed mean of zero
B. A key defining its numerical scale
C. A probability of one for each leaf
D. A compulsory normal curve

Source: T14-17, T44

Q030 Q1=10 and Q3=18. Find the upper inner outlier fence. Give the value to 4 decimal places.

A. 30.0000
B. 26.0000
C. 42.0000
D. -2.0000

Source: T14-17, T44

Q031 Q1=4 and Q3=12. Find the upper inner outlier fence. Give the value to 4 decimal places.

A. 20.0000
B. 36.0000
C. -8.0000
D. 24.0000

Source: T14-17, T44

Q032 Q1=20 and Q3=30. Find the upper inner outlier fence. Give the value to 4 decimal places.

A. 60.0000
B. 5.0000
C. 45.0000
D. 40.0000

Source: T14-17, T44

Fill in the gaps

F012 The distance from Q1 to Q3 is the _____ range.

Source: T14-17, T44

F013 A stem-and-leaf display requires a _____ to show its numerical scale.

Source: T14-17, T44

Topic 07 • Arithmetic, harmonic and geometric means

Source pathway: T18-21

Q033 For grouped frequency data, the estimated mean uses what as each class value?

A. Class width alone
B. Class midpoint
C. Cumulative count
D. Upper class label only

Source: T18-21

Q034 Which mean is suited to speeds over equal distances?

A. Harmonic mean
B. Unweighted arithmetic mean in all cases
C. Midrange
D. Mode

Source: T18-21

Q035 Which statement holds for positive observations?

A. Geometric mean always equals median
B. Harmonic mean always equals maximum
C. Arithmetic mean is always smallest
D. Harmonic mean ≤ geometric mean ≤ arithmetic mean

Source: T18-21

Q036 Adding 5 to every observation has what effect on the arithmetic mean?

A. It is multiplied by 5
B. It stays unchanged
C. It increases by 5
D. It increases by 25

Source: T18-21

Q037 Find the arithmetic mean of 3, 7, 4, 8, 5, 10. Give the value to 4 decimal places.

A. 5.1667
B. 6.1667
C. 37.0000
D. 7.1667

Source: T18-19

Q038 Find the arithmetic mean of 2, 3, 3, 4, 28. Give the value to 4 decimal places.

A. 8.0000
B. 40.0000
C. 9.0000
D. 7.0000

Source: T18-19

Q039 Find the arithmetic mean of 4, 8, 12, 16. Give the value to 4 decimal places.

A. 11.0000
B. 40.0000
C. 9.0000
D. 10.0000

Source: T18-19

Q040 Use x=0,1,2,3,4,5,6,7,8 with f=4,3,2,5,2,2,1,3,2 from T18. What is the checked mean? Give the value to 4 decimal places.

A. 9.2222
B. 0.2892
C. 3.4583
D. 6.9167

Source: T18-19

Q041 A vehicle covers two equal distances at speeds 30 and 60. Find the average speed. Give the value to 4 decimal places.

A. 90.0000
B. 40.0000
C. 45.0000
D. 42.4264

Source: T20

Q042 A vehicle covers two equal distances at speeds 40 and 80. Find the average speed. Give the value to 4 decimal places.

A. 53.3333
B. 120.0000
C. 60.0000
D. 56.5685

Source: T20

Fill in the gaps

F014 The reciprocal of the mean of reciprocals is the _____ mean.

Source: T18-21

F015 For positive data, the nth root of their product is the _____ mean.

Source: T18-21

F016 The corrected T18-19 numerator ∑fx is _____.

Source: T18-21

Topic 08 • Median, mode and robust location

Source pathway: T21-25

Q043 Before finding an ungrouped median, what should be done?

A. Delete duplicates
B. Replace every value by a midpoint
C. Square all values
D. Order the observations

Source: T21-25

Q044 The mode is which value?

A. Always the average
B. Always the largest value
C. A value having the greatest frequency
D. Always unique

Source: T21-25

Q045 For an even number of observations, the median is what?

A. Always the lower central value
B. The average of the two central ordered values
C. The mean of all values
D. The most frequent value

Source: T21-25

Q046 What does a grouped median formula primarily do?

A. Interpolate within the median class
B. Find a population size
C. Prove symmetry
D. Recover every original observation

Source: T21-25

Q047 Find the median of the ordered values 2, 4, 6, 8, 10, 12. Give the value to 4 decimal places.

A. 8.0000
B. 6.0000
C. 11.0000
D. 7.0000

Source: T21-22

Q048 Find the median of the ordered values 1, 4, 8, 10, 15. Give the value to 4 decimal places.

A. 12.0000
B. 7.0000
C. 8.0000
D. 9.0000

Source: T21-22

Q049 Find the median of the ordered values 3, 3, 4, 5, 18, 20. Give the value to 4 decimal places.

A. 5.5000
B. 4.5000
C. 8.5000
D. 3.5000

Source: T21-22

Q050 Using N=45, L=49.5, preceding cumulative count=15, median-class frequency=11 and width=10, estimate the median. Give the value to 4 decimal places.

A. 56.3182
B. 51.5455
C. 63.1364
D. 55.5000

Source: T22-23

Q051 For a modal class with L=49.5, width=10 and frequencies previous=8, modal=11, next=9, estimate the mode. Give the value to 4 decimal places.

A. 49.5000
B. 54.5000
C. 56.3000
D. 55.5000

Source: T24-25

Fill in the gaps

F017 The most frequent value is a _____.

Source: T21-25

F018 The central ordered value is the _____ for an odd sample size.

Source: T21-25

F019 The median of 2,4,6,8,10,12 is _____.

Source: T21-25

Topic 09 • Quartiles, deciles and percentiles

Source pathway: T25-28

Q052 The 7th decile equals which percentile?

A. 25th
B. 75th
C. 70th
D. 7th

Source: T25-28

Q053 The semi-interquartile range is what?

A. The mean divided by two
B. Half the interquartile range
C. Twice the range
D. The upper quartile alone

Source: T25-28

Q054 In the stated ungrouped interpolation rule, which position is used for probability p?

A. (n+1)p
B. p/n
C. (n-1)/p
D. n+p

Source: T25-28

Q055 Which count replaces N/2 when finding a general grouped percentile p?

A. N-p
B. N/p
C. 2N
D. pN

Source: T25-28

Q056 Use rank (n+1)p for sorted data 2,4,6,8,10,12,14. Find percentile p=0.25. Give the value to 4 decimal places.

A. 2.0000
B. 3.0000
C. 4.0000
D. 7.0000

Source: T25-27

Q057 Use rank (n+1)p for sorted data 2,4,6,8,10,12,14. Find percentile p=0.75. Give the value to 4 decimal places.

A. 11.0000
B. 12.0000
C. 15.0000
D. 6.0000

Source: T25-27

Q058 Q1=1.055 and Q3=1.230 on the source numerical scale. Find the IQR. Give the value to 4 decimal places.

A. 0.1750
B. 0.0875
C. 2.2850
D. 0.3500

Source: T26-27

Q059 For grouped data, p=0.25, N=45, L=39.5, C=7, f=8, h=10. Estimate Q1. Give the value to 4 decimal places.

A. 56.3182
B. 39.5000
C. 45.1250
D. 44.8125

Source: T12, T25-28

Fill in the gaps

F020 The 50th percentile is the _____.

Source: T25-28

F021 The third quartile is the _____ percentile.

Source: T25-28

F022 If Q1=4 and Q3=12, the semi-IQR is _____.

Source: T25-28

Topic 10 • Dispersion, variance and relative variation

Source pathway: T28-34

Q060 Which divisor is used for the usual unbiased sample variance?

A. n squared
B. The mean
C. n-1
D. n+1

Source: T28-34

Q061 Variance is expressed in what units?

A. Percentage points only
B. Squared measurement units
C. Original measurement units
D. No units in every case

Source: T28-34

Q062 The coefficient of variation is commonly computed as what?

A. 100 times SD divided by the mean
B. 100 times mean divided by SD
C. Variance minus mean
D. Range divided by median only

Source: T28-34

Q063 Adding a constant to all observations changes variance how?

A. It becomes zero
B. It increases by the constant
C. It is multiplied by the constant
D. It remains unchanged

Source: T28-34

Q064 Treat 3, 5, 7, 9, 11 as the entire population. Find its variance. Give the value to 4 decimal places.

A. 2.8284
B. 40.0000
C. 8.0000
D. 10.0000

Source: T29-32

Q065 Treat 2, 4, 6 as the entire population. Find its variance. Give the value to 4 decimal places.

A. 1.6330
B. 2.6667
C. 8.0000
D. 4.0000

Source: T29-32

Q066 Treat 1, 2, 3, 4 as the entire population. Find its variance. Give the value to 4 decimal places.

A. 1.2500
B. 5.0000
C. 1.1180
D. 1.6667

Source: T29-32

Q067 For a positive ratio-scale variable, mean=20 and SD=5. Find CV in percent. Give the value to 4 decimal places.

A. 400.0000
B. 0.2500
C. 12.5000
D. 25.0000

Source: T34

Q068 For a positive ratio-scale variable, mean=60 and SD=12. Find CV in percent. Give the value to 4 decimal places.

A. 500.0000
B. 0.2000
C. 20.0000
D. 10.0000

Source: T34

Q069 For a positive ratio-scale variable, mean=15 and SD=3. Find CV in percent. Give the value to 4 decimal places.

A. 500.0000
B. 20.0000
C. 0.2000
D. 10.0000

Source: T34

Fill in the gaps

F023 The positive square root of variance is the _____.

Source: T28-34

F024 The usual sample-variance denominator is _____.

Source: T28-34

F025 For the whole population 3,5,7,9,11, variance is _____.

Source: T28-34

Topic 11 • Skewness, kurtosis and summary output

Source pathway: T34-42

Q070 A distribution with a long right tail is described as what?

A. Positively skewed
B. Necessarily uniform
C. Necessarily symmetric
D. Negatively skewed

Source: T34-42

Q071 Normal-distribution excess kurtosis equals what?

A. 2
B. 3
C. 1
D. 0

Source: T34-42

Q072 Pearson's second skewness coefficient is which expression?

A. (mean-mode)/variance
B. SD/mean
C. 3(mean-median)/SD
D. Median minus range

Source: T34-42

Q073 Which claim about a symmetric distribution is valid?

A. Its mean must be zero
B. Its centre need not be zero
C. Its variance must be one
D. Every observation equals its mean

Source: T34-42

Q074 Mean=20, median=18, SD=4. Find Pearson second skewness. Give the value to 4 decimal places.

A. 1.5000
B. 4.5000
C. 2.5000
D. 0.5000

Source: T34-35

Q075 Mean=12, median=14, SD=3. Find Pearson second skewness. Give the value to 4 decimal places.

A. 1.0000
B. -3.0000
C. -1.0000
D. -2.0000

Source: T34-35

Q076 Mean=30, median=30, SD=5. Find Pearson second skewness. Give the value to 4 decimal places.

A. 1.0000
B. 3.0000
C. 0.0000
D. -1.0000

Source: T34-35

Q077 If ordinary kurtosis β2=5.4, what is excess kurtosis? Give the value to 4 decimal places.

A. 8.4000
B. 2.4000
C. 5.4000
D. -2.4000

Source: T35-36

Fill in the gaps

F026 A long right tail indicates _____ skewness.

Source: T34-42

F027 Normal-distribution kurtosis, before subtracting three, equals _____.

Source: T34-42

STA 112

Topic 12 • Random variables and probability models

Source pathway: T48-51, S12, S15

Q078 A random variable is best defined as what?

A. A real-valued function of outcomes
B. The sample space itself
C. A fixed population size only
D. An event with no numerical mapping

Source: T48-51, S12, S15

Q079 For a continuous variable with a density, P(X=c) equals what?

A. The density height at c
B. 1 for every c
C. The mean
D. 0

Source: T48-51, S12, S15

Q080 Which condition must a probability mass function satisfy?

A. Its mean equals 0
B. Every mass equals 1
C. Its nonnegative masses sum to 1
D. It must have six values

Source: T48-51, S12, S15

Q081 Why are two-dice sums not equally likely?

A. All dice are biased
B. Different sums arise from different numbers of ordered pairs
C. Every sum has exactly one outcome
D. The support is empty

Source: T48-51, S12, S15

Q082 A fair coin is tossed independently 2 times. What is the probability of at least one head? Give the value to 4 decimal places.

A. 0.7500
B. 0.3750
C. 0.2500
D. 0.5000

Source: T49-51

Q083 A fair coin is tossed independently 3 times. What is the probability of at least one head? Give the value to 4 decimal places.

A. 0.5000
B. 0.1250
C. 0.4375
D. 0.8750

Source: T49-51

Q084 A fair coin is tossed independently 4 times. What is the probability of at least one head? Give the value to 4 decimal places.

A. 0.4688
B. 0.0625
C. 0.9375
D. 0.5000

Source: T49-51

Fill in the gaps

F028 The support of X is the set of its possible _____.

Source: T48-51, S12, S15

F029 For a continuous density model, P(X=c) is _____.

Source: T48-51, S12, S15

Topic 13 • Expectation, variance and transformations

Source pathway: T52-55, S15-17, S21-26

Q085 Which expression gives variance from raw moments?

A. $E(X^2)-E(X)$
B. $E(X^2)-[E(X)]^2$
C. $[E(X)]^2-E(X^2)$
D. $E(X)-1$

Source: T52-55, S15-17, S21-26

Q086 If Y=a+bX, Var(Y) is what?

A. $b^2Var(X)$
B. $a^2Var(X)$
C. $a+bVar(X)$
D. $bVar(X)$

Source: T52-55, S15-17, S21-26

Q087 For independent X and Y, Var(X-Y) equals what?

A. Var(X)-Var(Y)
B. Var(X)Var(Y)
C. 0 always
D. Var(X)+Var(Y)

Source: T52-55, S15-17, S21-26

Q088 Which statement about expectation is correct?

A. Expectations add only under independence
B. $E(X^2)$ always equals $[E(X)]^2$
C. $E(X+Y)=E(X)+E(Y)$ without requiring independence
D. Expected value must be observed

Source: T52-55, S15-17, S21-26

Q089 E(X)=3, Var(X)=1. If Y=4+(3)X, find E(Y). Give the value to 4 decimal places.

A. 12.0000
B. 13.0000
C. 14.0000
D. 9.0000

Source: T52-54, S16-17

Q090 E(X)=3, Var(X)=1. If Y=4+(3)X, find Var(Y). Give the value to 4 decimal places.

A. 9.0000
B. 3.0000
C. 10.0000
D. 15.0000

Source: T52-54, S16-17

Q091 E(X)=5, Var(X)=4. If Y=10+(-2)X, find E(Y). Give the value to 4 decimal places.

A. 16.0000
B. 1.0000
C. -1.0000
D. 0.0000

Source: T52-54, S16-17

Q092 E(X)=5, Var(X)=4. If Y=10+(-2)X, find Var(Y). Give the value to 4 decimal places.

A. 8.0000
B. 28.0000
C. 16.0000
D. 17.0000

Source: T52-54, S16-17

Q093 E(X)=2, Var(X)=3. If Y=0+(5)X, find E(Y). Give the value to 4 decimal places.

A. 11.0000
B. 10.0000
C. 9.0000
D. 75.0000

Source: T52-54, S16-17

Q094 E(X)=2, Var(X)=3. If Y=0+(5)X, find Var(Y). Give the value to 4 decimal places.

A. 75.0000
B. 15.0000
C. 77.0000
D. 76.0000

Source: T52-54, S16-17

Q095 Two fair dice are independent. X is the smaller number shown. Find E(X). Give the value to 4 decimal places.

A. 3.5000
B. 7.0000
C. 2.0000
D. 2.5278

Source: T55 exercise 2

Q096 Two fair dice are independent. X is the smaller number shown. Find Var(X). Give the value to 4 decimal places.

A. 2.5278
B. 0.9857
C. 1.9715
D. 2.9167

Source: T55 exercise 2

Fill in the gaps

F030 Variance equals $E(X^2)$ minus the _____ of E(X).

Source: T52-55, S15-17, S21-26

F031 For a constant b, Var(bX) equals _____ times Var(X).

Source: T52-55, S15-17, S21-26

F032 For E(X)=3 and $E(X^2)=10$, Var(X)=_____.

Source: T52-55, S15-17, S21-26

Topic 14 • Events, axioms and probability laws

Source pathway: T56-61, S12, S27

Q097 Mutually exclusive events have what property?

A. They must be independent
B. Their intersection is empty
C. Their union is empty
D. Their probabilities are equal

Source: T56-61, S12, S27

Q098 Independent events satisfy which identity?

A. $P(A \cap B)=P(A)P(B)$
B. $P(A \cup B)=P(A)+P(B)$ always
C. $P(A \cap B)=0$ always
D. $P(A)=1-P(B)$ always

Source: T56-61, S12, S27

Q099 The general addition rule subtracts which term?

A. $P(A^c)$
B. $P(A)P(B)$ in every case
C. $P(S)$
D. $P(A \cap B)$

Source: T56-61, S12, S27

Q100 Two disjoint events each having positive probability are what?

A. Identical events
B. Always independent
C. Dependent
D. Both certain

Source: T56-61, S12, S27

Q101 $P(A)=0.4$, $P(B)=0.5$, $P(A \cap B)=0.2$. Find $P(A \cup B)$. Give the value to 4 decimal places.

A. 0.2000
B. 0.7000
C. 0.3000
D. 0.9000

Source: T60-61

Q102 $P(A)=0.6$, $P(B)=0.3$, $P(A \cap B)=0.1$. Find $P(A \cup B)$. Give the value to 4 decimal places.

A. 0.8000
B. 0.9000
C. 0.2000
D. 0.1000

Source: T60-61

Q103 $P(A)=0.7$, $P(B)=0.6$, $P(A \cap B)=0.5$. Find $P(A \cup B)$. Give the value to 4 decimal places.

A. 0.5000
B. 0.2000
C. 1.3000
D. 0.8000

Source: T60-61

Fill in the gaps

F033 The event of both A and B is their _____.

Source: T56-61, S12, S27

F034 Events that cannot occur together are mutually _____.

Source: T56-61, S12, S27

F035 If P(A)=0.4, P(B)=0.5 and P(A∩B)=0.2, P(A∪B)=_____.

Source: T56-61, S12, S27

Topic 15 • Conditional probability, tables and replacement

Source pathway: T62-72

Q104 The denominator of P(A given B) is what?

A. 1-P(B)
B. P(A∪B)
C. P(B)
D. P(A)

Source: T62-72

Q105 Without replacement, successive category draws are generally what?

A. Automatically independent
B. Dependent
C. Impossible to model
D. Always mutually exclusive as events

Source: T62-72

Q106 Which table total is used for an unconditional joint probability?

A. Grand total
B. Only the first row
C. Only the smallest cell
D. The difference of column totals

Source: T62-72

Q107 Which method reverses a conditional probability using prior group probabilities?

A. The harmonic mean
B. A class midpoint
C. A sample variance
D. Bayes rule

Source: T62-72

Q108 In a table of 100 units, a conditioning group contains 52, of whom 42 have the target property. Find the conditional probability. Give the value to 4 decimal places.

A. 0.5200
B. 0.4200
C. 0.8077
D. 0.1923

Source: T63

Q109 In a table of 100 units, a conditioning group contains 48, of whom 9 have the target property. Find the conditional probability. Give the value to 4 decimal places.

A. 0.4800
B. 0.1875
C. 0.8125
D. 0.0900

Source: T63

Q110 In a table of 100 units, a conditioning group contains 52, of whom 10 have the target property. Find the conditional probability. Give the value to 4 decimal places.

A. 0.1923
B. 0.5200
C. 0.1000
D. 0.8077

Source: T63

Q111 Wet season occupies 1/3 of a year and has rain probability 3/4. Dry season occupies 2/3 and has rain probability 1/6. Given rain on a random day, find P(wet season given rain). Give the value to 4 decimal places.

A. 0.7500
B. 0.3611
C. 0.3333
D. 0.6923

Source: T72

Q112 Three cards are drawn successively without replacement from a standard 52-card deck. Find the probability all three are aces. Give the value to 6 decimal places.

A. 0.000455
B. 0.000754
C. 0.000181
D. 0.005917

Source: T65, T68

Fill in the gaps

F036 In P(A given B), the conditioning event is _____.

Source: T62-72

F037 Sampling without replacement generally makes draws _____.

Source: T62-72

Topic 16 • Combinatorial analysis

Source pathway: S13-14, S66

Q113 Selecting an unordered committee uses which count?

A. Permutation in every case
B. Combination
C. Sample variance
D. Geometric density

Source: S13-14, S66

Q114 Arranging r distinct objects chosen from n without repetition uses what?

A. n!/(n-r)!
B. n!/[r!(n-r)!]
C. n+r
D. n/r

Source: S13-14, S66

Q115 What is 0 factorial?

A. Infinity
B. 0
C. Undefined in all counting formulas
D. 1

Source: S13-14, S66

Q116 For n distinct people around an unlabelled circle with reflections distinct, the count is what?

A. n!
B. 2n
C. (n-1)!
D. n^2

Source: S13-14, S66

Q117 Choose an unordered committee with exactly 3 men and 2 women from 6 men and 4 women. How many committees?

A. 130
B. 120
C. 240
D. 252

Source: S14, S66

Q118 Choose an unordered committee with exactly 2 men and 1 women from 5 men and 3 women. How many committees?

A. 30
B. 40
C. 56
D. 60

Source: S14, S66

Q119 Choose an unordered committee with exactly 2 men and 2 women from 7 men and 5 women. How many committees?

A. 220
B. 420
C. 495
D. 210

Source: S14, S66

Q120 How many distinct arrangements of all letters in ADELEKE are possible?

A. 5040
B. 2520
C. 840
D. 420

Source: S13

Q121 Using 15 distinct symbols, how many five-character passwords are possible with repetition allowed and no restriction on the first character?

A. 360360
B. 759375
C. 50625
D. 75

Source: S66

Fill in the gaps

F038 An unordered selection is counted with a _____.

Source: S13-14, S66

F039 For counting identities, 0! equals _____.

Source: S13-14, S66

F040 A five-person committee from ten distinct people has _____ possible memberships.

Source: S13-14, S66

Topic 17 • Discrete cumulative distribution functions

Source pathway: S17-18; T49-50, T74-77

Q122 A CDF F(t) equals which probability?

A. P(X≤t)
B. E(X)
C. P(X>t)
D. P(X=t) in every model

Source: S17-18; T49-50, T74-77

Q123 At a discrete point x, its probability mass equals what?

A. The entire CDF height at x
B. The derivative at every point
C. The mean
D. The CDF jump at x

Source: S17-18; T49-50, T74-77

Q124 Every CDF must be what?

A. Strictly decreasing
B. Always differentiable
C. Nondecreasing and right-continuous
D. Always a straight line

Source: S17-18; T49-50, T74-77

Q125 For any CDF, P(a<X≤b) equals what?

A. F(a)+F(b)
B. F(b)-F(a)
C. 1-F(b)
D. F(a)-F(b)

Source: S17-18; T49-50, T74-77

Q126 X has masses 1/4,1/2,1/4 at -1,2,5. What is F(2)? Give the value to 4 decimal places.

A. 0.7500
B. 0.2500
C. 1.0000
D. 0.5000

Source: S17-18

Q127 A CDF jumps from 1/6 to 1/2 at x=6. Find P(X=6). Give the value to 4 decimal places.

A. 0.6667
B. 0.1667
C. 0.5000
D. 0.3333

Source: S18

Q128 X has masses 1/4,1/2,1/4 at -1,2,5. Find P(X>2). Give the value to 4 decimal places.

A. 0.7500
B. 0.5000
C. 0.2500
D. 1.0000

Source: S17-18

Fill in the gaps

F041 A discrete CDF is right-_____ at its jumps.

Source: S17-18; T49-50, T74-77

F042 The probability mass at x is the size of the CDF _____ there.

Source: S17-18; T49-50, T74-77

Topic 18 • Continuous densities and cumulative probabilities

Source pathway: T51, T54-55, S18-20, S22-26

Q129 A valid continuous density must have which total area?

A. Its mean
B. 1
C. Its maximum height
D. 0

Source: T51, T54-55, S18-20, S22-26

Q130 Which quantity is integrated to find an interval probability?

A. The probability density
B. Only the support endpoints
C. The variance
D. The CDF a second time in every case

Source: T51, T54-55, S18-20, S22-26

Q131 Can a density height exceed 1?

A. No, every density height is a probability
B. Only for invalid models
C. Only if variance is negative
D. Yes, if its total area is still 1

Source: T51, T54-55, S18-20, S22-26

Q132 Where a continuous CDF is differentiable, its derivative is what?

A. The variance
B. The support size
C. The density
D. The mean

Source: T51, T54-55, S18-20, S22-26

Q133 $f(x)=k(4x-2x^2)$ on $0<x<2$ and zero elsewhere. Find k. Give the value to 4 decimal places.

A. 0.5000
B. 0.3750
C. 2.6667
D. 0.7500

Source: S19

Q134 $f(x)=x^2/9$ on $0<x<3$ and zero elsewhere. Find $P(1<X<2)$. Give the value to 4 decimal places.

A. 0.2593
B. 0.0370
C. 0.7778
D. 0.2963

Source: S20

Q135 $f(x)=81x^2$ on $0<x<1/3$ and zero elsewhere. Find E(X). Give the value to 4 decimal places.

A. 0.7500
B. 0.3333
C. 0.1111
D. 0.2500

Source: S22

Fill in the gaps

F043 A continuous density integrates to _____ over its full support.

Source: T51, T54-55, S18-20, S22-26

F044 Differentiating a differentiable CDF gives the _____.

Source: T51, T54-55, S18-20, S22-26

Topic 19 • Bernoulli and binomial distributions

Source pathway: T74-84, S28-35, S41

Q136 Which condition is required for a binomial model?

A. Sampling without replacement in every case
B. A continuous response
C. Independent trials with constant success probability
D. A random number of trials in every case

Source: T74-84, S28-35, S41

Q137 A Bernoulli variable has which support?

A. All real numbers
B. 0 and 1
C. Only 1 and 2
D. All positive integers

Source: T74-84, S28-35, S41

Q138 The mean of Bin(n,p) is what?

A. np
B. n+p
C. p/n
D. n/p

Source: T74-84, S28-35, S41

Q139 At least r successes means which event?

A. X=r only
B. X≤r
C. X>r
D. X≥r

Source: T74-84, S28-35, S41

Q140 X~Bin(5,0.5). Find P(X=2). Give the value to 4 decimal places.

A. 0.4125
B. 0.6875
C. 0.3125
D. 0.1562

Source: T77-83, S33-35

Q141 X~Bin(6,0.5). Find P(X=2). Give the value to 4 decimal places.

A. 0.1172
B. 0.2344
C. 0.7656
D. 0.3344

Source: T77-83, S33-35

Q142 X~Bin(8,0.25). Find P(X=0). Give the value to 4 decimal places.

A. 0.1001
B. 0.0501
C. 0.8999
D. 0.2001

Source: T77-83, S33-35

Q143 X~Bin(4,0.3). Find P(X=2). Give the value to 4 decimal places.

A. 0.3646
B. 0.7354
C. 0.1323
D. 0.2646

Source: T77-83, S33-35

Q144 X~Bin(10,0.2). Find P(X=1). Give the value to 4 decimal places.

A. 0.1342
B. 0.7316
C. 0.2684
D. 0.3684

Source: T77-83, S33-35

Q145 For X~Bin(14,0.2), find P(X≤2), correcting the source arithmetic. Give the value to 4 decimal places.

A. 0.6722
B. 0.4481
C. 0.1539
D. 0.1298

Source: T80-83

Q146 A fair coin is tossed independently six times. Find P(at least five heads). Give the value to 4 decimal places.

A. 0.1094
B. 0.0938
C. 0.0156
D. 0.2344

Source: T78-80

Fill in the gaps

F045 A single 0/1 success indicator has a _____ distribution.

Source: T74-84, S28-35, S41

F046 The number of successes in n independent equal-p trials is _____.

Source: T74-84, S28-35, S41

F047 Bin(8,0.25) has mean _____.

Source: T74-84, S28-35, S41

Topic 20 • Poisson counts and approximation

Source pathway: T85-86, S31, S35-37

Q147 A Poisson distribution has which relation between mean and variance?

A. Mean is zero
B. Variance is zero
C. Variance equals λ^2 always
D. Both equal λ

Source: T85-86, S31, S35-37

Q148 The support of a Poisson count is what?

A. All real numbers
B. Only 0 and 1
C. 0,1,2,...
D. 1,2,3,... only

Source: T85-86, S31, S35-37

Q149 Which binomial setting is most suitable for a Poisson approximation?

A. Any continuous density
B. Large n and small p with $\lambda=np$
C. Large n alone regardless of p
D. A single certain success

Source: T85-86, S31, S35-37

Q150 Under a constant-rate Poisson process, doubling exposure does what to λ?

A. Doubles it
B. Leaves it unchanged
C. Squares it
D. Halves it

Source: T85-86, S31, S35-37

Q151 X~Poisson(2). Find P(X=0). Give the value to 4 decimal places.

A. 0.1853
B. 0.8647
C. 0.0677
D. 0.1353

Source: T85-86, S35-37

Q152 X~Poisson(3). Find P(X=2). Give the value to 4 decimal places.

A. 0.7760
B. 0.1120
C. 0.2240
D. 0.2740

Source: T85-86, S35-37

Q153 X~Poisson(1.6). Find P(X=3). Give the value to 4 decimal places.

A. 0.0689
B. 0.1378
C. 0.1878
D. 0.8622

Source: T85-86, S35-37

Q154 X~Poisson(7.5). Find P(X=6). Give the value to 4 decimal places.

A. 0.1367
B. 0.8633
C. 0.0684
D. 0.1867

Source: T85-86, S35-37

Q155 X~Poisson(7.5). Find P(X≤8), using the at-most wording in the source question. Give the value to 4 decimal places.

A. 0.1321
B. 0.3380
C. 0.4750
D. 0.6620

Source: T85-86

Fill in the gaps

F048 For a Poisson variable, both mean and variance equal _____.

Source: T85-86, S31, S35-37

F049 A rare-event binomial approximation commonly uses Poisson parameter λ=_____.

Source: T85-86, S31, S35-37

F050 Poisson(3) has variance _____.

Source: T85-86, S31, S35-37

Topic 21 • Geometric waiting times

Source pathway: T87, S37-40, S44-45

Q156 Here a geometric random variable counts what?

A. Successes in a fixed number of trials
B. Only failures including no final success
C. Trials through the first success
D. Events in a fixed time for every model

Source: T87, S37-40, S44-45

Q157 For that geometric convention, the mean is what?

A. 1-p
B. 1/p
C. p
D. p^2

Source: T87, S37-40, S44-45

Q158 Which property is associated with a geometric waiting time?

A. Memorylessness
B. A fixed maximum of five trials
C. Negative variance
D. A continuous uniform density

Source: T87, S37-40, S44-45

Q159 Changing from trials to failures before success changes variance how?

A. It becomes zero
B. It is divided by p
C. It decreases by 1
D. It stays the same

Source: T87, S37-40, S44-45

Q160 Independent trials have success probability 0.25. Find the probability that the first success occurs on trial 3. Give the value to 4 decimal places.

A. 0.2500
B. 0.1055
C. 0.1406
D. 0.8594

Source: T87, S37-40

Q161 Independent trials have success probability 0.2. Find the probability that the first success occurs on trial 4. Give the value to 4 decimal places.

A. 0.8976
B. 0.1024
C. 0.2000
D. 0.0819

Source: T87, S37-40

Q162 Independent trials have success probability 0.5. Find the probability that the first success occurs on trial 2. Give the value to 4 decimal places.

A. 0.2500
B. 0.5000
C. 0.7500
D. 0.1250

Source: T87, S37-40

Q163 A geometric trial count has p=0.25. Find its variance. Give the value to 4 decimal places.

A. 0.1875
B. 3.0000
C. 4.0000
D. 12.0000

Source: T87, S38-40

Fill in the gaps

F051 The trial-count geometric support begins at _____.

Source: T87, S37-40, S44-45

F052 The geometric waiting-time model has the _____ property.

Source: T87, S37-40, S44-45

F053 A trial-count geometric variable with p=0.2 has mean _____.

Source: T87, S37-40, S44-45

Topic 22 • Hypergeometric sampling without replacement

Source pathway: T87-89

Q164 Which model counts successes in a fixed-size SRS without replacement?

A. Normal point mass
B. Geometric trial count
C. Hypergeometric
D. Binomial in every finite population

Source: T87-89

Q165 In a hypergeometric model, what does K denote?

A. Observed mean
B. Population number of successes
C. Sample size
D. Sample number of failures

Source: T87-89

Q166 Which factor reduces hypergeometric variance relative to np(1-p)?

A. (N-n)/(N-1)
B. N+n
C. n^2
D. (N-1)/(N-n)

Source: T87-89

Q167 If the entire finite population is sampled, the success-count variance is what?

A. N
B. K
C. 1
D. 0

Source: T87-89

Q168 An SRS of 10 is drawn without replacement from 50 units, of which 4 are successes. Find P(exactly 2 successes). Give the value to 4 decimal places.

A. 0.2524
B. 0.8476
C. 0.1524
D. 0.0762

Source: T87-89

Q169 An SRS of 4 is drawn without replacement from 20 units, of which 5 are successes. Find P(exactly 1 successes). Give the value to 4 decimal places.

A. 0.5304
B. 0.4696
C. 0.5696
D. 0.2348

Source: T87-89

Q170 An SRS of 3 is drawn without replacement from 12 units, of which 4 are successes. Find P(exactly 2 successes). Give the value to 4 decimal places.

A. 0.2182
B. 0.1091
C. 0.7818
D. 0.3182

Source: T87-89

Fill in the gaps

F054 A finite-population count sampled without replacement has a _____ distribution.

Source: T87-89

F055 When all population units are sampled, count variance is _____.

Source: T87-89

Topic 23 • Uniform or rectangular distribution

Source pathway: T89-90, S40-41

Q171 The density of Uniform(a,b) inside its interval is what?

A. (a+b)/2
B. b-a
C. 1/a in every case
D. 1/(b-a)

Source: T89-90, S40-41

Q172 The mean of Uniform(a,b) is what?

A. ab
B. b-a
C. (a+b)/2
D. $(b-a)^2/12$

Source: T89-90, S40-41

Q173 Equal-length intervals inside a uniform support have what?

A. Probability 1 each
B. Equal probabilities
C. Zero probability always
D. Equal individual outcomes

Source: T89-90, S40-41

Q174 The variance of Uniform(a,b) equals what?

A. $(b-a)^2/12$
B. 1/(b-a)
C. $(a+b)^2$
D. (b-a)/2

Source: T89-90, S40-41

Q175 X~Uniform(0,4). Find P(3<X<4). Give the value to 4 decimal places.

A. 0.4000
B. 0.1250
C. 0.5500
D. 0.2500

Source: T89-90, S40-41

Q176 X~Uniform(2,8). Find P(3<X<5). Give the value to 4 decimal places.

A. 0.6333
B. 0.1667
C. 0.3333
D. 0.4833

Source: T89-90, S40-41

Q177 X~Uniform(10,20). Find P(12<X<17). Give the value to 4 decimal places.

A. 0.2500
B. 0.5000
C. 0.6500
D. 0.8000

Source: T89-90, S40-41

Fill in the gaps

F056 The uniform density height on (a,b) is _____.

Source: T89-90, S40-41

F057 Uniform(a,b) has mean _____.

Source: T89-90, S40-41

Topic 24 • Normal distributions and standard scores

Source pathway: T90-94

Q178 A standard normal distribution has which parameters?

A. Mean 0 and variance 1
B. Mean 0 and variance 0
C. Any mean and variance 1
D. Mean 1 and variance 0

Source: T90-94

Q179 Which expression standardises $X \sim N(\mu,\sigma^2)$?

A. X/μ
B. $X-\sigma$
C. $(X-\mu)/\sigma^2$
D. $(X-\mu)/\sigma$

Source: T90-94

Q180 Which area equals $P(Z>z)$?

A. $\Phi(z)$
B. $\Phi(z)-1$
C. $1-\Phi(z)$
D. $z\Phi(z)$

Source: T90-94

Q181 For a normal distribution, mean, median and mode are what?

A. All zero regardless of μ
B. Equal to μ
C. Always different
D. All equal to σ

Source: T90-94

Q182 X is normal with mean 300 and SD 50. Find P(X>362). Give the value to 4 decimal places.

A. 0.1075
B. 0.0537
C. 0.2075
D. 0.8925

Source: T92-94

Q183 X is normal with mean 50 and SD 10. Find P(X>60). Give the value to 4 decimal places.

A. 0.0793
B. 0.2587
C. 0.8413
D. 0.1587

Source: T92-94

Q184 X is normal with mean 65 and SD 15. Find P(X>80). Give the value to 4 decimal places.

A. 0.2587
B. 0.8413
C. 0.1587
D. 0.0793

Source: T92-94

Fill in the gaps

F058 A standard normal variable has variance _____.

Source: T90-94

F059 Subtracting μ and dividing by σ is called _____.

Source: T90-94

Topic 25 • Gamma integrals and distribution

Source pathway: S42-44; T74 objective only

Q185 For gamma shape r and rate λ, the mean is what?

A. rλ
B. r/λ
C. r/λ²
D. λ/r

Source: S42-44; T74 objective only

Q186 For a positive integer m, Γ(m) equals what?

A. (m-1)!
B. m!
C. 1/m
D. m²

Source: S42-44; T74 objective only

Q187 Which quantity is the scale corresponding to rate λ?

A. λ+r
B. λ²
C. rλ
D. 1/λ

Source: S42-44; T74 objective only

Q188 Setting gamma shape r=1 gives which distribution?

A. Uniform on all real numbers
B. Bernoulli
C. Exponential
D. Hypergeometric

Source: S42-44; T74 objective only

Q189 X is gamma with shape 3 and rate 2. Find Var(X). Give the value to 4 decimal places.

A. 12.0000
B. 0.7500
C. 1.5000
D. 0.6667

Source: S42-44

Q190 X is gamma with shape 4 and rate 5. Find Var(X). Give the value to 4 decimal places.

A. 0.1600
B. 0.8000
C. 100.0000
D. 1.2500

Source: S42-44

Q191 X is gamma with shape 2 and rate 3. Find Var(X). Give the value to 4 decimal places.

A. 0.6667
B. 1.5000
C. 18.0000
D. 0.2222

Source: S42-44

Q192 Evaluate the integral of $x^2 \exp(-5x)$ from 0 to infinity. Give the value to 4 decimal places.

A. 0.0800
B. 0.0480
C. 0.0160
D. 0.0080

Source: S42

Fill in the gaps

F060 Gamma rate and scale are _____ of each other.

Source: S42-44; T74 objective only

F061 For a positive integer m, Γ(m) equals _____.

Source: S42-44; T74 objective only

F062 Gamma shape 3 and rate 2 has mean _____.

Source: S42-44; T74 objective only

Topic 26 • Supplement bridge: descriptive revision

Source pathway: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q193 How should the early STA 111 labels inside the supplement be treated in this pack?

A. Reassign textbook chapters arbitrarily
B. Preserve them as source evidence while following the user's course allocation
C. Pretend they read STA 112
D. Delete all descriptive material

Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q194 Which calculation best checks a handwritten frequency total?

A. Sum the individual frequencies independently
B. Count class widths
C. Trust the final underline only
D. Use the largest frequency

Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q195 What should be done with conflicting quartile conventions?

A. Ignore ordering
B. Mix them until answers match
C. Use only the maximum
D. State one convention and apply it consistently

Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q196 What is the proper treatment of cropped formula fragments?

A. Assume a new course heading
B. Treat every fragment as an instruction
C. Flag them and avoid inventing missing symbols
D. Complete them from guesswork as a quotation

Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q197 For the handwritten values 8,7,4,6,7,2,3, what is the mean? Give the value to 4 decimal places.

A. 5.3000
B. 5.2857
C. 7.0000
D. 6.0000

Source: S7

Q198 Recompute S5 using frequencies 20,41,52,14,49,98,19,15,31,65,18,92,66,70 and class midpoints 5.5,15.5,...,135.5. Find the mean. Give the value to 4 decimal places.

A. 79.6385
B. 79.4992
C. 85.9839
D. 69.0200

Source: S5-8

Fill in the gaps

F063 The supplement contains early pages visibly labelled STA _____.

Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

F064 Handwritten frequency totals should be checked by independent _____.

Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Topic 27 • Supplement bridge: sampling and standard error

Source pathway: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q199 A sampling distribution describes what?

A. A list of questionnaire questions
B. Only the population labels
C. Only the individual raw scores
D. A statistic over repeated samples under a design

Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q200 The standard error of a mean for independent observations with SD σ is what?

A. σ/n
B. n/σ
C. σ/√n
D. σ√n

Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q201 Quota selection differs from stratified probability sampling mainly in what?

A. Its group counts can never match proportions
B. Selection within groups need not be random
C. It must survey everyone
D. It requires no population definition

Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q202 Which sequence is necessary in a systematic probability sample?

A. Random start followed by the stated fixed interval
B. Convenient start followed by arbitrary steps
C. Only the largest labels
D. The same person every time

Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q203 Independent observations have population SD 12. For a sample mean with n=36, find its standard error. Give the value to 4 decimal places.

A. 12.0000
B. 0.3333
C. 72.0000
D. 2.0000

Source: S48, S54

Q204 Independent observations have population SD 20. For a sample mean with n=100, find its standard error. Give the value to 4 decimal places.

A. 0.2000
B. 20.0000
C. 2.0000
D. 200.0000

Source: S48, S54

Q205 Independent observations have population SD 8. For a sample mean with n=16, find its standard error. Give the value to 4 decimal places.

A. 0.5000
B. 2.0000
C. 8.0000
D. 32.0000

Source: S48, S54

Fill in the gaps

F065 The SD of a statistic over repeated samples is its standard _____.

Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

F066 Proportionate stratified allocation uses nh=nNh/_____.

Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Topic 28 • Supplement bridge: testing and paired comparisons

Source pathway: S49-61; cross-reference STA 121 topics 36-39

Q206 Which data structure calls for a paired t test?

A. Before and after readings on the same units
B. A four-category goodness-of-fit table
C. A single categorical count
D. Two unrelated random groups in every case

Source: S49-61; cross-reference STA 121 topics 36-39

Q207 For n independent paired differences, t-test degrees of freedom are what?

A. n^2
B. 2n-2
C. n+1
D. n-1

Source: S49-61; cross-reference STA 121 topics 36-39

Q208 The direction of a test is determined by what?

A. The largest sample observation
B. The sign that yields significance after looking
C. The prespecified alternative hypothesis
D. The number of decimal places

Source: S49-61; cross-reference STA 121 topics 36-39

Q209 What does failing to reject H0 imply?

A. The sample was necessarily random
B. The test did not provide sufficient evidence against H0
C. H0 is proven true
D. The effect is exactly zero

Source: S49-61; cross-reference STA 121 topics 36-39

Q210 Independent paired differences are 2,4,3,5,1. Find the t statistic for H0: mean difference=0. Give the value to 4 decimal places.

A. 4.2426
B. 3.0000
C. 6.0000
D. 1.3416

Source: S55

Fill in the gaps

F067 A paired t test uses within-pair _____.

Source: S49-61; cross-reference STA 121 topics 36-39

F068 A nondirectional alternative uses the symbol _____.

Source: S49-61; cross-reference STA 121 topics 36-39

Topic 29 • Analysis of variance and post-hoc comparisons

Source pathway: S60, S62-63, S65

Q211 The one-way ANOVA alternative states what?

A. Every observation differs
B. All sample variances are zero
C. All means must differ
D. At least one population mean differs

Source: S60, S62-63, S65

Q212 The ANOVA F statistic is what?

A. The sum of group means
B. Grand mean divided by variance
C. Between-group mean square divided by error mean square
D. Error SS divided by total sample size only

Source: S60, S62-63, S65

Q213 With k groups and N observations, error degrees of freedom are what?

A. k-1
B. N-k
C. N-1
D. Nk

Source: S60, S62-63, S65

Q214 What is the role of a post-hoc procedure after an omnibus comparison?

A. Identify particular group differences while addressing multiple comparisons
B. Prove all observations are normal
C. Create a random sample
D. Replace the data by ranks automatically

Source: S60, S62-63, S65

Q215 ANOVA has N=22 observations in k=4 groups. Find error degrees of freedom.

A. 22
B. 21
C. 3
D. 18

Source: S62

Q216 Groups are (2,4), (5,7), (8,10). Find between-group sum of squares. Give the value to 4 decimal places.

A. 6.0000
B. 18.0000
C. 36.0000
D. 42.0000

Source: S62, S65

Q217 For the same three groups, SSB=36, SSE=6, N=6 and k=3. Find F. Give the value to 4 decimal places.

A. 18.0000
B. 9.0000
C. 3.0000
D. 6.0000

Source: S62, S65

Fill in the gaps

F069 The ANOVA ratio is MS between divided by MS _____.

Source: S60, S62-63, S65

F070 ANOVA error degrees of freedom are N minus _____.

Source: S60, S62-63, S65

F071 For N=22 and k=4 in one-way ANOVA, error df=_____.

Source: S60, S62-63, S65

Topic 30 • Chi-square goodness of fit and independence

Source pathway: S55, S63-65

Q218 A goodness-of-fit chi-square test concerns what?

A. Observed category counts versus specified probabilities
B. Two unknown means only
C. A single continuous score without categories
D. The ordering of passwords

Source: S55, S63-65

Q219 Under independence, a cell's expected count equals what?

A. Row total plus column total
B. Grand total divided by observed cell count
C. The smaller margin
D. Row total times column total divided by grand total

Source: S55, S63-65

Q220 For an r by c independence table, degrees of freedom equal what?

A. rc
B. rc-1 in every case
C. (r-1)(c-1)
D. r+c

Source: S55, S63-65

Q221 A significant chi-square independence result establishes what?

A. Equal category counts
B. Evidence of association under the model
C. Equal population means
D. Causation by itself

Source: S55, S63-65

Q222 Four specified equally likely categories have observed counts 18,22,20,20. Find χ^2. Give the value to 4 decimal places.

A. 0.4000
B. 0.2000
C. 4.0000
D. 8.0000

Source: S55, S63-64

Q223 For an independence test in a 3 by 4 table, find degrees of freedom.

A. 11
B. 7
C. 12
D. 6

Source: S55, S63-64

Fill in the gaps

F072 A chi-square statistic sums squared O-E differences divided by _____.

Source: S55, S63-65

F073 Independence-test expected counts use row total times column total divided by _____ total.

Source: S55, S63-65

Topic 31 • Experimental design and correlation concepts

Source pathway: S65; S60, S62 contextual mentions

Q224 Which design principle assigns treatments by chance?

A. Replication alone
B. Convenience recruitment
C. Randomisation
D. Local control alone

Source: S65; S60, S62 contextual mentions

Q225 Which principle provides independent units receiving the same treatment?

A. Repeated formatting
B. Replication
C. Cumulative frequency
D. An identifier code

Source: S65; S60, S62 contextual mentions

Q226 Blocking is an example of which design principle?

A. Local control
B. Random sampling by definition
C. Purely descriptive averaging
D. A probability density

Source: S65; S60, S62 contextual mentions

Q227 An observational correlation alone supports which conclusion?

A. No need for a population definition
B. Certain causation
C. Random treatment assignment
D. Association, with possible confounding

Source: S65; S60, S62 contextual mentions

Fill in the gaps

F074 Assigning treatments by chance is _____.

Source: S65; S60, S62 contextual mentions

F075 Blocking is a form of local _____.

Source: S65; S60, S62 contextual mentions

STA 121

Topic 32 • Sampling foundations and simple random sampling

Source pathway: T95-98

Q228 A sampling frame is what?

A. The sample variance
B. The population mean
C. The operational list or structure used to select units
D. A confidence interval

Source: T95-98

Q229 Which property defines an SRS of fixed size without replacement?

A. The sample must be homogeneous
B. Every possible sample of that size is equally likely
C. Every unit is selected
D. Only the first unit is random

Source: T95-98

Q230 When using random labels without replacement, what should happen to duplicates?

A. Reject the repeated selection and continue by the rule
B. Keep every duplicate as a new unit
C. Replace the population mean
D. Always stop immediately

Source: T95-98

Q231 Which statement about large samples is correct?

A. They remove the need for a frame
B. They always eliminate bias
C. They make every design an SRS
D. They do not automatically remove selection bias

Source: T95-98

Q232 How many distinct unordered SRS samples of size 3 can be selected without replacement from 10 units?

A. 720
B. 1000
C. 120
D. 30

Source: T96-98

Q233 There are 20 labelled units and an SRS of size 5 without replacement. What is one unit's inclusion probability? Give the value to 4 decimal places.

A. 0.7500
B. 0.2500
C. 0.2000
D. 0.0500

Source: T96-98

Fill in the gaps

F076 The operational list for selecting a sample is a sampling _____.

Source: T95-98

F077 A fixed-size SRS gives every possible sample an equal _____.

Source: T95-98

Topic 33 • Stratified and cluster sampling

Source pathway: T98-100

Q234 Proportionate stratified allocation assigns nh as what?

A. nNh/N
B. NNh/n
C. Nh+n
D. n/Nh

Source: T98-100

Q235 In one-stage cluster sampling, after selecting clusters one observes what?

A. Some units from every possible cluster necessarily
B. Only one fixed unit overall
C. Only the largest stratum
D. All units in the chosen clusters

Source: T98-100

Q236 Strata must be what?

A. All the same size
B. Overlapping by definition
C. Mutually exclusive and collectively cover the population
D. Chosen only after seeing outcomes

Source: T98-100

Q237 Why can cluster sampling be statistically less efficient than SRS?

A. All clusters are normal distributions
B. Units within clusters can be similar
C. It always uses larger n
D. It always has zero cost

Source: T98-100

Q238 Use proportionate stratified allocation with N=500, stratum size=140, total sample=50. Find the stratum sample size.

A. 14
B. 19
C. 9
D. 140

Source: T98-103

Q239 Use proportionate stratified allocation with N=600, stratum size=200, total sample=60. Find the stratum sample size.

A. 25
B. 15
C. 200
D. 20

Source: T98-103

Q240 Use proportionate stratified allocation with N=1200, stratum size=420, total sample=200. Find the stratum sample size.

A. 65
B. 75
C. 70
D. 420

Source: T98-103

Fill in the gaps

F078 In stratified sampling, a sample is drawn from _____ stratum.

Source: T98-100

F079 In one-stage cluster sampling, all units in selected _____ are observed.

Source: T98-100

Topic 34 • Systematic and multistage sampling

Source pathway: T101-103

Q241 For the elementary systematic design with N divisible by n, the interval is what?

A. N-n
B. N/n
C. n/N
D. N+n

Source: T101-103

Q242 Which frame feature can make a systematic interval problematic?

A. Periodicity aligned with the interval
B. Unique valid labels
C. A known total size
D. A documented random start

Source: T101-103

Q243 Selecting schools, then classes, then students is which design?

A. A frequency polygon
B. Purposive sampling necessarily
C. One-stage census only
D. Multistage sampling

Source: T101-103

Q244 In a multistage design, equal school probabilities guarantee equal student probabilities when what?

A. Always, regardless of school size
B. Only if no students are sampled
C. Further size and selection conditions make them equal
D. Never under any design

Source: T101-103

Q245 A systematic sample has N=100, n=10, starting position 5. What is the fourth selected position?

A. 25
B. 35
C. 15
D. 45

Source: T101-102

Q246 A systematic sample has N=60, n=10, starting position 2. What is the fourth selected position?

A. 20
B. 8
C. 26
D. 14

Source: T101-102

Q247 A systematic sample has N=120, n=12, starting position 7. What is the fourth selected position?

A. 17
B. 27
C. 47
D. 37

Source: T101-102

Fill in the gaps

F080 The elementary systematic sampling interval is N divided by _____.

Source: T101-103

F081 Sampling schools then students is a _____ design.

Source: T101-103

Topic 35 • Quota and purposive sampling

Source pathway: T102-103

Q248 A nonrandom selection that fills fixed subgroup targets is what?

A. A binomial density
B. SRS by definition
C. Quota sampling
D. A complete census

Source: T102-103

Q249 Deliberately choosing experts meeting study criteria is what?

A. A random-number lottery
B. Purposive sampling
C. Systematic probability sampling
D. Uniform distribution

Source: T102-103

Q250 Which claim is justified by balanced quota counts alone?

A. Only that the chosen quotas were filled
B. Every unit had known selection probability
C. An SRS confidence level is guaranteed
D. The sample is unbiased for every variable

Source: T102-103

Q251 Which aim most directly calls for a probability design?

A. Choosing only available friends
B. Exploring a few specialist experiences only
C. Selecting a single illustrative case
D. Population estimation with design-based uncertainty

Source: T102-103

Q252 A quota sample of 30 is to include 70% in a subgroup. How many places are assigned to it?

A. 26
B. 16
C. 21
D. 9

Source: T102-103

Q253 A quota sample of 50 is to include 40% in a subgroup. How many places are assigned to it?

A. 30
B. 20
C. 15
D. 25

Source: T102-103

Fill in the gaps

F082 Nonrandom filling of subgroup targets is _____ sampling.

Source: T102-103

F083 Selection based deliberately on relevant expertise is _____ sampling.

Source: T102-103

Topic 36 • Hypotheses, errors and decisions

Source pathway: T104-108

Q254 A Type I error occurs when what happens?

A. A true H0 is rejected
B. The sample size grows
C. A sample mean is calculated
D. A false H0 is not rejected

Source: T104-108

Q255 Test power equals what?

A. α+β
B. 1-α in every situation
C. The sample mean
D. 1-β

Source: T104-108

Q256 A p-value is calculated assuming what?

A. H1 is certainly true
B. The observed difference is impossible
C. H0 and the test model
D. H0 has a known posterior probability

Source: T104-108

Q257 For a two-sided z test at 5%, rejection occurs when what holds?

A. $z<1.96$ always
B. Absolute z exceeds 1.96
C. Absolute z below 1.645
D. $z>1.645$ only

Source: T104-108

Q258 At a specified alternative, Type II error probability β=0.2. Find power. Give the value to 4 decimal places.

A. 0.8000
B. 0.2000
C. 0.4000
D. 0.9000

Source: T108

Q259 At a specified alternative, Type II error probability β=0.35. Find power. Give the value to 4 decimal places.

A. 0.3250
B. 0.8250
C. 0.3500
D. 0.6500

Source: T108

Q260 At a specified alternative, Type II error probability β=0.1. Find power. Give the value to 4 decimal places.

A. 0.9500
B. 0.1000
C. 0.9000
D. 0.4500

Source: T108

Fill in the gaps

F084 Rejecting a true H0 is a Type _____ error.

Source: T104-108

F085 Power equals one minus _____.

Source: T104-108

F086 If β=0.2, test power is _____.

Source: T104-108

Topic 37 • One-sample tests for a population mean

Source pathway: T108-113

Q261 For a normal population with unknown SD, a one-sample mean test uses what?

A. χ^2=(mean-μ0)/n
B. t=(mean-μ0)/(s/√n)
C. z=(mean-μ0)/s^2
D. F=mean/s

Source: T108-113

Q262 If population σ is known, the SE of the sample mean is what?

A. σ/√n
B. σ^2√n
C. σn
D. σ/n^2

Source: T108-113

Q263 For a left-tailed t test, evidence against H0 is what?

A. A t near zero
B. A large sample mean in the opposite direction
C. Any positive t
D. A sufficiently negative t

Source: T108-113

Q264 Which statement about n=30 is sound?

A. Every n below 30 is invalid
B. All large samples are normal populations
C. It is a classroom heuristic, not the definition of t versus z
D. Every n above 30 has known σ

Source: T108-113

Q265 For a one-sample t statistic, n=10, sample mean=2.24, sample SD=0.36, null mean=2. Find the signed statistic. Give the value to 4 decimal places.

A. 1.1082
B. 2.1082
C. -2.1082
D. 3.1082

Source: T109-112

Q266 For a one-sample t statistic, n=25, sample mean=54, sample SD=10, null mean=50. Find the signed statistic. Give the value to 4 decimal places.

A. 2.0000
B. 3.0000
C. -2.0000
D. 1.0000

Source: T109-112

Q267 For a one-sample t statistic, n=16, sample mean=18, sample SD=4, null mean=20. Find the signed statistic. Give the value to 4 decimal places.

A. -3.0000
B. -1.0000
C. 2.0000
D. -2.0000

Source: T109-112

Fill in the gaps

F087 For a one-sample t test, degrees of freedom are _____.

Source: T108-113

F088 A left-tailed alternative seeks a sufficiently _____ signed statistic.

Source: T108-113

Topic 38 • Tests comparing two population means

Source pathway: T113-116

Q268 A pooled two-sample t procedure assumes what about population variances?

A. They are necessarily unequal
B. Both are known to be zero
C. They are equal
D. They must equal the means

Source: T113-116

Q269 Independent pooled t-test degrees of freedom equal what?

A. n1n2-1
B. n1+n2-2
C. n1-n2
D. n1+n2

Source: T113-116

Q270 Which data should instead be analysed through within-pair differences?

A. Matched before-and-after data
B. Two separately sampled unrelated groups
C. Two independent category totals only
D. A single Poisson count

Source: T113-116

Q271 The standard error for independent known-variance mean differences is what?

A. $\sigma 1+\sigma 2$
B. The difference of variances
C. The product of sample means
D. Square root of $\sigma 1^2/n1+\sigma 2^2/n2$

Source: T113-116

Q272 Independent samples have means 54 and 50, n1=n2=25 and known SDs 5 and 5. Find z for a null difference of zero. Give the value to 4 decimal places.

A. 2.0000
B. 4.0000
C. 2.8284
D. 1.4142

Source: T113-114

Q273 Independent pooled samples have n1=8 and n2=6. Find t-test degrees of freedom.

A. 13
B. 12
C. 14
D. 6

Source: T114

Fill in the gaps

F089 A pooled t test assumes equal population _____.

Source: T113-116

F090 For two independent pooled samples, df=n1+n2-_____.

Source: T113-116

Topic 39 • Tests for one and two proportions

Source pathway: T116-120

Q274 For a one-proportion z test under H0:p=p0, use which variance factor?

A. p0(1-p0)
B. The observed mean score
C. The pooled mean of ages
D. phat(1-phat) in every null test

Source: T116-120

Q275 Under H0:p1=p2, the pooled proportion equals what?

A. (n1+n2)/(x1+x2)
B. (x1-x2)/(n1-n2)
C. phat1+phat2
D. (x1+x2)/(n1+n2)

Source: T116-120

Q276 For an interval on p1-p2, which SE is normally used in the elementary formula?

A. Zero
B. A sample SD of raw ages
C. Unpooled estimated SE
D. The null pooled SE automatically

Source: T116-120

Q277 A difference from 0.03 to 0.08 is what size in percentage points?

A. 0.05
B. 5
C. 0.5
D. 50

Source: T116-120

Q278 There are 65 successes in 100 independent trials. Under H0:p=0.60, find the one-proportion z statistic. Give the value to 4 decimal places.

A. 1.0206
B. 0.0500
C. 1.9600
D. 1.0483

Source: T118-119

Q279 Pool 30 successes out of 1000 with 100 successes out of 1200 under equality of proportions. Find the pooled proportion. Give the value to 4 decimal places.

A. 0.0300
B. 0.0833
C. 0.0567
D. 0.0591

Source: T117, T120

Q280 A random sample has 600 category successes out of 800. Under H0:p=0.80, find the signed one-proportion z statistic. Give the value to 4 decimal places.

A. -1.9600
B. -3.2660
C. -3.5355
D. 3.5355

Source: T120 exercise 5

Fill in the gaps

F091 In a one-proportion null test, the standard error uses the _____ proportion p0.

Source: T116-120

F092 Under equality of two proportions, successes are pooled across both _____.

Source: T116-120

Topic 40 • Point estimation and properties of estimators

Source pathway: T121-124

Q281 An estimator is what?

A. Always the fixed population parameter
B. A rule applied to random sample data
C. Only one observed number
D. A guaranteed correct answer

Source: T121-124

Q282 Unbiasedness means what?

A. The estimator's expectation equals the parameter
B. Variance is always zero
C. Every sample estimate equals the parameter
D. The estimator is always positive

Source: T121-124

Q283 Consistency concerns what?

A. The same printed font
B. Equal estimates in every sample
C. A fixed bias of one
D. Convergence to the parameter as sample size grows

Source: T121-124

Q284 Among unbiased estimators of the same parameter, smaller variance indicates what?

A. A larger fixed parameter
B. No possible information
C. Greater efficiency in that comparison
D. Greater bias necessarily

Source: T121-124

Q285 An unbiased estimator T has variance 9. Another unbiased estimator U has variance 4. Find Var(T)/Var(U). Give the value to 4 decimal places.

A. 13.0000
B. 2.2500
C. 0.4444
D. 5.0000

Source: T123

Fill in the gaps

F093 An unbiased estimator has expectation equal to the _____.

Source: T121-124

F094 Convergence to the parameter as n grows is _____.

Source: T121-124

Topic 41 • Confidence intervals for means and proportions

Source pathway: T125-128, T132-133

Q286 A confidence interval's margin of error equals what?

A. Critical multiplier times SE
B. Sample count times variance
C. The confidence level alone
D. Twice the mean

Source: T125-128, T132-133

Q287 A 95% frequentist confidence level describes what?

A. A 95% probability assigned to H0
B. A 95% probability that the fixed parameter changes
C. The percentage of data inside every interval
D. Long-run coverage of the interval procedure

Source: T125-128, T132-133

Q288 For the same data, increasing confidence generally does what to interval width?

A. Makes it negative
B. Leaves it exactly fixed
C. Increases it
D. Decreases it

Source: T125-128, T132-133

Q289 When σ is unknown and a normal small-sample mean model is appropriate, use which multiplier?

A. A Poisson mean
B. A t critical value with n-1 df
C. Always 1
D. The sample maximum

Source: T125-128, T132-133

Q290 A mean interval uses known σ=10, n=100, sample mean=50 and z=1.96. Find its lower endpoint. Give the value to 4 decimal places.

A. 48.0400
B. 1.9600
C. 46.0800
D. 51.9600

Source: T125-127

Q291 A mean interval uses known σ=8, n=64, sample mean=30 and z=1.96. Find its lower endpoint. Give the value to 4 decimal places.

A. 1.9600
B. 26.0800
C. 31.9600
D. 28.0400

Source: T125-127

Q292 A mean interval uses known σ=6, n=36, sample mean=20 and z=1.96. Find its lower endpoint. Give the value to 4 decimal places.

A. 16.0800
B. 1.9600
C. 18.0400
D. 21.9600

Source: T125-127

Q293 For independent proportions 30/1000 and 100/1200, find the upper endpoint of the 95% Wald interval for p1-p2, using z=1.96. Give the value to 4 decimal places.

A. -0.0533
B. -0.0345
C. 0.0189
D. -0.0722

Source: T133 exercise 7

Fill in the gaps

F095 Interval width equals _____ times its margin of error.

Source: T125-128, T132-133

F096 A confidence level describes repeated-sampling _____.

Source: T125-128, T132-133

F097 A margin of error of 2 gives interval width _____.

Source: T125-128, T132-133

Topic 42 • Sample size and precision planning

Source pathway: T128-133

Q294 Required sample sizes from precision formulas should be rounded how?

A. Up to the next integer
B. Always down
C. To zero if fractional
D. To the nearest tenth

Source: T128-133

Q295 With no prior proportion, which planning value is conservative?

A. 0.99 in every case
B. 0
C. 1
D. 0.5

Source: T128-133

Q296 Under mean precision planning, halving the margin requires what approximate sample-size change?

A. Divide by 2
B. Multiply by 2
C. Multiply by 4
D. Leave n unchanged

Source: T128-133

Q297 Which issue is not automatically repaired by increasing planned n?

A. Finite arithmetic precision
B. Selection bias
C. A wide SE from small n under the model
D. Random sampling variability under the model

Source: T128-133

Q298 Plan a mean interval with z=1.96, known/planning SD=6.95 and margin E=1. Find the minimum integer n.

A. 186
B. 14
C. 187
D. 185

Source: T129-131

Q299 Plan a mean interval with z=1.96, known/planning SD=10 and margin E=2. Find the minimum integer n.

A. 96
B. 10
C. 98
D. 97

Source: T129-131

Q300 Plan a mean interval with z=1.96, known/planning SD=5 and margin E=1. Find the minimum integer n.

A. 98
B. 96
C. 97
D. 10

Source: T129-131

Fill in the gaps

F098 Precision sample-size calculations are rounded _____.

Source: T128-133

F099 The conservative unknown-proportion planning value is _____.

Source: T128-133

F100 Using z=1.96, p*=0.5 and E=0.05, minimum planned n=_____.

Source: T128-133

Part II • Keys and worked answers

Compact MCQ key

Question / answer	Question / answer	Question / answer	Question / answer	Question / answer
Q001 B	Q002 A	Q003 D	Q004 C	Q005 B
Q006 A	Q007 D	Q008 C	Q009 B	Q010 A
Q011 D	Q012 C	Q013 B	Q014 A	Q015 D
Q016 C	Q017 B	Q018 A	Q019 D	Q020 C
Q021 B	Q022 A	Q023 D	Q024 C	Q025 B
Q026 A	Q027 D	Q028 C	Q029 B	Q030 A
Q031 D	Q032 C	Q033 B	Q034 A	Q035 D
Q036 C	Q037 B	Q038 A	Q039 D	Q040 C
Q041 B	Q042 A	Q043 D	Q044 C	Q045 B
Q046 A	Q047 D	Q048 C	Q049 B	Q050 A
Q051 D	Q052 C	Q053 B	Q054 A	Q055 D
Q056 C	Q057 B	Q058 A	Q059 D	Q060 C
Q061 B	Q062 A	Q063 D	Q064 C	Q065 B
Q066 A	Q067 D	Q068 C	Q069 B	Q070 A
Q071 D	Q072 C	Q073 B	Q074 A	Q075 D
Q076 C	Q077 B	Q078 A	Q079 D	Q080 C
Q081 B	Q082 A	Q083 D	Q084 C	Q085 B
Q086 A	Q087 D	Q088 C	Q089 B	Q090 A
Q091 D	Q092 C	Q093 B	Q094 A	Q095 D
Q096 C	Q097 B	Q098 A	Q099 D	Q100 C
Q101 B	Q102 A	Q103 D	Q104 C	Q105 B
Q106 A	Q107 D	Q108 C	Q109 B	Q110 A
Q111 D	Q112 C	Q113 B	Q114 A	Q115 D
Q116 C	Q117 B	Q118 A	Q119 D	Q120 C
Q121 B	Q122 A	Q123 D	Q124 C	Q125 B
Q126 A	Q127 D	Q128 C	Q129 B	Q130 A
Q131 D	Q132 C	Q133 B	Q134 A	Q135 D

Question / answer	Question / answer	Question / answer	Question / answer	Question / answer
Q136 C	Q137 B	Q138 A	Q139 D	Q140 C
Q141 B	Q142 A	Q143 D	Q144 C	Q145 B
Q146 A	Q147 D	Q148 C	Q149 B	Q150 A
Q151 D	Q152 C	Q153 B	Q154 A	Q155 D
Q156 C	Q157 B	Q158 A	Q159 D	Q160 C
Q161 B	Q162 A	Q163 D	Q164 C	Q165 B
Q166 A	Q167 D	Q168 C	Q169 B	Q170 A
Q171 D	Q172 C	Q173 B	Q174 A	Q175 D
Q176 C	Q177 B	Q178 A	Q179 D	Q180 C
Q181 B	Q182 A	Q183 D	Q184 C	Q185 B
Q186 A	Q187 D	Q188 C	Q189 B	Q190 A
Q191 D	Q192 C	Q193 B	Q194 A	Q195 D
Q196 C	Q197 B	Q198 A	Q199 D	Q200 C
Q201 B	Q202 A	Q203 D	Q204 C	Q205 B
Q206 A	Q207 D	Q208 C	Q209 B	Q210 A
Q211 D	Q212 C	Q213 B	Q214 A	Q215 D
Q216 C	Q217 B	Q218 A	Q219 D	Q220 C
Q221 B	Q222 A	Q223 D	Q224 C	Q225 B
Q226 A	Q227 D	Q228 C	Q229 B	Q230 A
Q231 D	Q232 C	Q233 B	Q234 A	Q235 D
Q236 C	Q237 B	Q238 A	Q239 D	Q240 C
Q241 B	Q242 A	Q243 D	Q244 C	Q245 B
Q246 A	Q247 D	Q248 C	Q249 B	Q250 A
Q251 D	Q252 C	Q253 B	Q254 A	Q255 D
Q256 C	Q257 B	Q258 A	Q259 D	Q260 C
Q261 B	Q262 A	Q263 D	Q264 C	Q265 B
Q266 A	Q267 D	Q268 C	Q269 B	Q270 A
Q271 D	Q272 C	Q273 B	Q274 A	Q275 D
Q276 C	Q277 B	Q278 A	Q279 D	Q280 C
Q281 B	Q282 A	Q283 D	Q284 C	Q285 B

Question / answer	Question / answer	Question / answer	Question / answer	Question / answer
Q286 A	Q287 D	Q288 C	Q289 B	Q290 A
Q291 D	Q292 C	Q293 B	Q294 A	Q295 D
Q296 C	Q297 B	Q298 A	Q299 D	Q300 C

Worked MCQ answers

STA 111 • worked answers

Topic 01 • Statistical thinking and inference

Q001 • B - Descriptive statistics

It summarises the available observations without necessarily generalising beyond them.

Original question on source concepts | Source: T1-4

Q002 • A - Inductive

Induction moves from sample evidence to a population conclusion with uncertainty.

Original question on source concepts | Source: T1-4

Q003 • D - Parameter

A parameter is a characteristic of the specified population.

Original question on source concepts | Source: T1-4

Q004 • C - Define the question and target population

A clear purpose and population determine which data and methods are relevant.

Original question on source concepts | Source: T1-4

Topic 02 • Variables and the meaning of data

Q005 • B - A categorical identifier

Its digits identify a record; arithmetic differences between identifiers do not measure a quantity.

Original question on source concepts | Source: T4-6, T9

Q006 • A - Exact elapsed time

Elapsed time can take values throughout intervals in a continuous model.

Original question on source concepts | Source: T4-6, T9

Q007 • D - It has countably many possible values

Discrete support can be finite or countably infinite and need not consist only of integers.

Original question on source concepts | Source: T4-6, T9

Q008 • C - Discrete quantitative

Completed years form a countable set, even though underlying elapsed age can be continuous.

Original question on source concepts | Source: T4-6, T9

Topic 03 • Data sources, collection and quality

Q009 • B - Secondary data

The table was collected earlier by another producer or for another purpose.

Original question on source concepts | Source: T5-8

Q010 • A - An experiment

Manipulation distinguishes an experimental approach from passive observation.

Original question on source concepts | Source: T5-8

Q011 • D - Response depends on the characteristic being studied

Systematic response differences can make respondents unrepresentative of those selected.

Original question on source concepts | Source: T5-8

Q012 • C - A moderated focus group

Focus groups are a qualitative collection method for perceptions and experiences.

Original question on source concepts | Source: T5-8

Topic 04 • Official statistics and responsible use

Q013 • B - Central Bank of Nigeria

T8-9 lists the CBN among producers of financial statistics; this is a source example.

Original question on source concepts | Source: T8-9

Q014 • A - Its denominator or reference group

A percentage is meaningful only relative to a specified whole.

Original question on source concepts | Source: T8-9

Q015 • D - To make the time coverage clear

Users need to know when the reported observations apply.

Original question on source concepts | Source: T8-9

Q016 • C - Government, researchers, businesses and the public

The source describes a broad set of users for planning and interpretation.

Original question on source concepts | Source: T8-9

Topic 05 • Frequency tables and statistical graphs

Q017 • B - f/N

Relative frequency is the proportion of all observations in that class.

Original question on source concepts | Source: T10-15, T43-47

Q018 • A - Upper class boundaries

The accumulated count through each class belongs at its upper boundary, with an initial zero point.

Original question on source concepts | Source: T10-15, T43-47

Q019 • D - Frequency divided by class width

Area equals height times width, so frequency density preserves the frequency area.

Original question on source concepts | Source: T10-15, T43-47

Q020 • C - Bar chart

Separated bars are appropriate for categories; a histogram displays quantitative intervals.

Original question on source concepts | Source: T10-15, T43-47

Q021 • B - 24.4444

Relative frequency = 100×11/45 = 24.444444%.

Adapted source calculation, independently recomputed | Source: T12-13

Q022 • A - 20.0000

Relative frequency = 100×8/40 = 20.000000%.

Original numerical application of source method | Source: T12-13

Q023 • D - 30.0000

Relative frequency = 100×18/60 = 30.000000%.

Original numerical application of source method | Source: T12-13

Q024 • C - 26

Add the first four counts: 3+4+8+11=26.

Adapted source calculation, independently recomputed | Source: T10-15, T43-47

Q025 • B - 34.5000

The midpoint is (30+39)/2=34.5.

Original numerical application of source method | Source: T12-13

Topic 06 • Stem-and-leaf displays and box plots

Q026 • A - Interquartile range

The box runs from Q1 to Q3, so its width is Q3-Q1.

Original question on source concepts | Source: T14-17, T44

Q027 • D - 1.5 times the IQR

The conventional inner fences are Q1-1.5 IQR and Q3+1.5 IQR.

Original question on source concepts | Source: T14-17, T44

Q028 • C - The most extreme observation within the fence

The whisker endpoint is an actual non-outlying observation under this convention.

Original question on source concepts | Source: T14-17, T44

Q029 • B - A key defining its numerical scale

A key resolves whether stem 2 and leaf 4 means 24, 2.4 or another scale.

Original question on source concepts | Source: T14-17, T44

Q030 • A - 30.0000

IQR=18-10=8. Upper fence=18+1.5×8=30.0.

Original numerical application of source method | Source: T14-17, T44

Q031 • D - 24.0000

IQR=12-4=8. Upper fence=12+1.5×8=24.0.

Original numerical application of source method | Source: T14-17, T44

Q032 • C - 45.0000

IQR=30-20=10. Upper fence=30+1.5×10=45.0.

Original numerical application of source method | Source: T14-17, T44

Topic 07 • Arithmetic, harmonic and geometric means

Q033 • B - Class midpoint

Midpoints approximate the unknown positions of observations within each class.

Original question on source concepts | Source: T18-21

Q034 • A - Harmonic mean

Total distance divided by total time leads to a harmonic mean for equal-distance speeds.

Original question on source concepts | Source: T18-21

Q035 • D - Harmonic mean ≤ geometric mean ≤ arithmetic mean

The standard ordering holds for positive values, with equality when all are equal.

Original question on source concepts | Source: T18-21

Q036 • C - It increases by 5

A location shift adds the same constant to the mean.

Original question on source concepts | Source: T18-21

Q037 • B - 6.1667

Sum=37, count=6; mean=37/6=6.166667.

Original numerical application of source method | Source: T18-19

Q038 • A - 8.0000

Sum=40, count=5; mean=40/5=8.000000.

Original numerical application of source method | Source: T18-19

Q039 • D - 10.0000

Sum=40, count=4; mean=40/4=10.000000.

Original numerical application of source method | Source: T18-19

Q040 • C - 3.4583

Products sum to 83 and frequencies to 24. Mean=83/24=3.458333; the printed numerator 166 is incorrect.

Adapted source calculation, independently recomputed | Source: T18-19

Q041 • B - 40.0000

Total-distance/total-time gives H=2/(1/30+1/60)=40.000000.

Original numerical application of source method | Source: T20

Q042 • A - 53.3333

Total-distance/total-time gives H=2/(1/40+1/80)=53.333333.

Original numerical application of source method | Source: T20

Topic 08 • Median, mode and robust location

Q043 • D - Order the observations

The median is defined by positions in ordered data.

Original question on source concepts | Source: T21-25

Q044 • C - A value having the greatest frequency

The most frequent value is a mode, and ties can give multiple modes.

Original question on source concepts | Source: T21-25

Q045 • B - The average of the two central ordered values

The standard elementary convention averages positions n/2 and n/2+1.

Original question on source concepts | Source: T21-25

Q046 • A - Interpolate within the median class

It estimates the median using cumulative counts and a within-class interpolation assumption.

Original question on source concepts | Source: T21-25

Q047 • D - 7.0000

With n=6, use the middle position or average the two middle positions. Median=7.0.

Original numerical application of source method | Source: T21-22

Q048 • C - 8.0000

With n=5, use the middle position or average the two middle positions. Median=8.

Original numerical application of source method | Source: T21-22

Q049 • B - 4.5000

With n=6, use the middle position or average the two middle positions. Median=4.5.

Original numerical application of source method | Source: T21-22

Q050 • A - 56.3182

Median=49.5+[(45/2-15)/11]×10=56.318182.

Adapted source calculation, independently recomputed | Source: T22-23

Q051 • D - 55.5000

d1=3, d2=2; mode=49.5+[3/(3+2)]×10=55.5.

Adapted source calculation, independently recomputed | Source: T24-25

Topic 09 • Quartiles, deciles and percentiles

Q052 • C - 70th

Decile j corresponds to percentile 10j.

Original question on source concepts | Source: T25-28

Q053 • B - Half the interquartile range

Semi-IQR=(Q3-Q1)/2.

Original question on source concepts | Source: T25-28

Q054 • A - (n+1)p

This pack explicitly uses the (n+1)p rank rule when the resulting rank is within the observed positions.

Original question on source concepts | Source: T25-28

Q055 • D - pN

The required cumulative count is the requested fraction p of the total N.

Original question on source concepts | Source: T25-28

Q056 • C - 4.0000

n=7, rank=8×0.25=2; the value at that position is 4.

Original numerical application of source method | Source: T25-27

Q057 • B - 12.0000

n=7, rank=8×0.75=6; the value at that position is 12.

Original numerical application of source method | Source: T25-27

Q058 • A - 0.1750

IQR=1.230-1.055=0.175. No unresolved physical unit is assumed.

Adapted source calculation, independently recomputed | Source: T26-27

Q059 • D - 44.8125

pN=11.25; Q1=39.5+[(11.25-7)/8]×10=44.8125.

Adapted source calculation, independently recomputed | Source: T12, T25-28

Topic 10 • Dispersion, variance and relative variation

Q060 • C - n-1

One degree of freedom is used when the sample mean is estimated from the same data.

Original question on source concepts | Source: T28-34

Q061 • B - Squared measurement units

Variance averages squared deviations; SD returns to original units.

Original question on source concepts | Source: T28-34

Q062 • A - 100 times SD divided by the mean

CV measures spread relative to a positive ratio-scale mean.

Original question on source concepts | Source: T28-34

Q063 • D - It remains unchanged

Differences from the shifted mean remain the same.

Original question on source concepts | Source: T28-34

Q064 • C - 8.0000

Mean=7; sum of squared deviations=40. Divide by N=5 to obtain 8.000000.

Adapted source calculation, independently recomputed | Source: T29-32

Q065 • B - 2.6667

Mean=4; sum of squared deviations=8. Divide by N=3 to obtain 2.666667.

Original numerical application of source method | Source: T29-32

Q066 • A - 1.2500

Mean=2.5; sum of squared deviations=5. Divide by N=4 to obtain 1.250000.

Original numerical application of source method | Source: T29-32

Q067 • D - 25.0000

CV=100×5/20=25%.

Original numerical application of source method | Source: T34

Q068 • C - 20.0000

CV=100×12/60=20%.

Original numerical application of source method | Source: T34

Q069 • B - 20.0000

CV=100×3/15=20%.

Original numerical application of source method | Source: T34

Topic 11 • Skewness, kurtosis and summary output

Q070 • A - Positively skewed

The sign refers to the direction of the long tail.

Original question on source concepts | Source: T34-42

Q071 • D - 0

Ordinary kurtosis is 3, so subtracting 3 gives excess kurtosis 0.

Original question on source concepts | Source: T34-42

Q072 • C - 3(mean-median)/SD

This is the location-based second Pearson formula, distinct from moment skewness.

Original question on source concepts | Source: T34-42

Q073 • B - Its centre need not be zero

Symmetry can occur around any location.

Original question on source concepts | Source: T34-42

Q074 • A - 1.5000

Use 3(mean-median)/SD = 3×(20-18)/4=1.5.

Original numerical application of source method | Source: T34-35

Q075 • D - -2.0000

Use 3(mean-median)/SD = 3×(12-14)/3=-2.

Original numerical application of source method | Source: T34-35

Q076 • C - 0.0000

Use 3(mean-median)/SD = 3×(30-30)/5=0.

Original numerical application of source method | Source: T34-35

Q077 • B - 2.4000

Excess kurtosis=β2-3=5.4-3=2.4.

Original numerical application of source method | Source: T35-36

STA 112 • worked answers

Topic 12 • Random variables and probability models

Q078 • A - A real-valued function of outcomes

It assigns a numerical value to each outcome of an experiment.

Original question on source concepts | Source: T48-51, S12, S15

Q079 • D - 0

A single point has zero area under a continuous density.

Original question on source concepts | Source: T48-51, S12, S15

Q080 • C - Its nonnegative masses sum to 1

Normalisation and nonnegativity make the masses a probability model.

Original question on source concepts | Source: T48-51, S12, S15

Q081 • B - Different sums arise from different numbers of ordered pairs

For fair independent dice, the 36 ordered pairs are equally likely, but their sums have unequal multiplicities.

Original question on source concepts | Source: T48-51, S12, S15

Q082 • A - 0.7500

Use the complement of all tails: 1-(1/2)^2=0.750000.

Original numerical application of source method | Source: T49-51

Q083 • D - 0.8750

Use the complement of all tails: 1-(1/2)^3=0.875000.

Original numerical application of source method | Source: T49-51

Q084 • C - 0.9375

Use the complement of all tails: 1-(1/2)^4=0.937500.

Original numerical application of source method | Source: T49-51

Topic 13 • Expectation, variance and transformations

Q085 • B - $E(X^2)-[E(X)]^2$

Variance is the second moment minus the square of the mean.

Original question on source concepts | Source: T52-55, S15-17, S21-26

Q086 • A - $b^2Var(X)$

Translation does not change variance and scaling squares the multiplier.

Original question on source concepts | Source: T52-55, S15-17, S21-26

Q087 • D - Var(X)+Var(Y)

Independence removes covariance; the negative coefficient is squared.

Original question on source concepts | Source: T52-55, S15-17, S21-26

Q088 • C - E(X+Y)=E(X)+E(Y) without requiring independence

Linearity of expectation does not need independence, when expectations exist.

Original question on source concepts | Source: T52-55, S15-17, S21-26

Q089 • B - 13.0000

Linearity gives E(Y)=4+(3)×3=13; variance is not used to calculate this mean.

Original numerical application of source method | Source: T52-54, S16-17

Q090 • A - 9.0000

Var(Y)=(3)²×1=9; adding 4 has no effect on variance.

Original numerical application of source method | Source: T52-54, S16-17

Q091 • D - 0.0000

Linearity gives E(Y)=10+(-2)×5=0; variance is not used to calculate this mean.

Original numerical application of source method | Source: T52-54, S16-17

Q092 • C - 16.0000

Var(Y)=(-2)²×4=16; adding 10 has no effect on variance.

Original numerical application of source method | Source: T52-54, S16-17

Q093 • B - 10.0000

Linearity gives E(Y)=0+(5)×2=10; variance is not used to calculate this mean.

Original numerical application of source method | Source: T52-54, S16-17

Q094 • A - 75.0000

Var(Y)=(5)²×3=75; adding 0 has no effect on variance.

Original numerical application of source method | Source: T52-54, S16-17

Q095 • D - 2.5278

For x=1,...,6, counts of ordered pairs with minimum x are 11,9,7,5,3,1. The weighted numerator is 91; divide by 36 to get 2.527778.

Adapted source calculation, independently recomputed | Source: T55 exercise 2

Q096 • C - 1.9715

The minimum has counts 11,9,7,5,3,1 out of 36. E(X)=91/36 and E(X²)=301/36.
Variance=301/36-(91/36)²=1.97145062.

Adapted source calculation, independently recomputed | Source: T55 exercise 2

Topic 14 • Events, axioms and probability laws

Q097 • B - Their intersection is empty

They cannot occur together in one outcome.

Original question on source concepts | Source: T56-61, S12, S27

Q098 • A - P(A∩B)=P(A)P(B)

This is the general product criterion for independence.

Original question on source concepts | Source: T56-61, S12, S27

Q099 • D - P(A∩B)

The overlap has been counted in both P(A) and P(B), so subtract it once.

Original question on source concepts | Source: T56-61, S12, S27

Q100 • C - Dependent

Their joint probability is zero but the product of their positive probabilities is not zero.

Original question on source concepts | Source: T56-61, S12, S27

Q101 • B - 0.7000

Addition rule: 0.4+0.5-0.2=0.700000.

Original numerical application of source method | Source: T60-61

Q102 • A - 0.8000

Addition rule: 0.6+0.3-0.1=0.800000.

Original numerical application of source method | Source: T60-61

Q103 • D - 0.8000

Addition rule: 0.7+0.6-0.5=0.800000.

Original numerical application of source method | Source: T60-61

Topic 15 • Conditional probability, tables and replacement

Q104 • C - P(B)

Conditioning restricts the reference group to B, assuming P(B)>0.

Original question on source concepts | Source: T62-72

Q105 • B - Dependent

The first draw changes the category composition available for the second.

Original question on source concepts | Source: T62-72

Q106 • A - Grand total

A joint event is measured relative to the full set of units.

Original question on source concepts | Source: T62-72

Q107 • D - Bayes rule

Bayes rule combines conditional likelihood and prior probability, then normalises by the event probability.

Original question on source concepts | Source: T62-72

Q108 • C - 0.8077

Restrict the denominator to the group: 42/52=0.807692.

Adapted source calculation, independently recomputed | Source: T63

Q109 • B - 0.1875

Restrict the denominator to the group: 9/48=0.187500.

Adapted source calculation, independently recomputed | Source: T63

Q110 • A - 0.1923

Restrict the denominator to the group: 10/52=0.192308.

Adapted source calculation, independently recomputed | Source: T63

Q111 • D - 0.6923

P(rain)=(1/3)(3/4)+(2/3)(1/6)=13/36. P(wet and rain)=1/4. Bayes gives (1/4)/(13/36)=9/13.

Adapted source calculation, independently recomputed | Source: T72

Q112 • C - 0.000181

(4/52)(3/51)(2/50)=1/5525≈0.00018100. Available aces and total cards both decrease.

Adapted source calculation, independently recomputed | Source: T65, T68

Topic 16 • Combinatorial analysis

Q113 • B - Combination

Changing the order of the same selected members does not create a different committee.

Original question on source concepts | Source: S13-14, S66

Q114 • A - n!/(n-r)!

An ordered selection counts each arrangement separately.

Original question on source concepts | Source: S13-14, S66

Q115 • D - 1

The convention 0!=1 makes factorial counting identities consistent.

Original question on source concepts | Source: S13-14, S66

Q116 • C - (n-1)!

Fixing one person's position removes rotational duplication.

Original question on source concepts | Source: S13-14, S66

Q117 • B - 120

$C(6,3)\times C(4,2)=20\times 6=120$.

Adapted source calculation, independently recomputed | Source: S14, S66

Q118 • A - 30

$C(5,2)\times C(3,1)=10\times 3=30$.

Original numerical application of source method | Source: S14, S66

Q119 • D - 210

$C(7,2)\times C(5,2)=21\times 10=210$.

Original numerical application of source method | Source: S14, S66

Q120 • C - 840

There are 7 letters with E repeated 3 times. Distinct arrangements=7!/3!=840.

Adapted source calculation, independently recomputed | Source: S13

Q121 • B - 759375

There are 15 choices independently at each of five positions: $15^5=759375$.

Adapted source calculation, independently recomputed | Source: S66

Topic 17 • Discrete cumulative distribution functions

Q122 • A - P(X≤t)

It accumulates probability up to and including t.

Original question on source concepts | Source: S17-18; T49-50, T74-77

Q123 • D - The CDF jump at x

The mass is F(x)-F(x-), rather than F(x) alone.

Original question on source concepts | Source: S17-18; T49-50, T74-77

Q124 • C - Nondecreasing and right-continuous

Accumulated probability cannot decrease, and the ≤ convention makes the CDF right-continuous.

Original question on source concepts | Source: S17-18; T49-50, T74-77

Q125 • B - F(b)-F(a)

Subtracting accumulated probability through a leaves probability above a through b.

Original question on source concepts | Source: S17-18; T49-50, T74-77

Q126 • A - 0.7500

CDF includes the mass at 2: 1/4+1/2=3/4.

Adapted source calculation, independently recomputed | Source: S17-18

Q127 • D - 0.3333

The mass is the jump 1/2-1/6=1/3.

Adapted source calculation, independently recomputed | Source: S18

Q128 • C - 0.2500

Only the mass at 5 remains above 2, so the probability is 1/4.

Adapted source calculation, independently recomputed | Source: S17-18

Topic 18 • Continuous densities and cumulative probabilities

Q129 • B - 1

Probability over the full support must be one.

Original question on source concepts | Source: T51, T54-55, S18-20, S22-26

Q130 • A - The probability density

An interval probability is area under the density over that interval.

Original question on source concepts | Source: T51, T54-55, S18-20, S22-26

Q131 • D - Yes, if its total area is still 1

Probability constrains areas, not the pointwise density height.

Original question on source concepts | Source: T51, T54-55, S18-20, S22-26

Q132 • C - The density

Differentiating the cumulative integral recovers the density.

Original question on source concepts | Source: T51, T54-55, S18-20, S22-26

Q133 • B - 0.3750

The integral of $4x-2x^2$ is 8/3 over (0,2). Thus $k\times 8/3=1$ and k=3/8.

Adapted source calculation, independently recomputed | Source: S19

Q134 • A - 0.2593

Integrate $x^2/9$: $[x^3/27]$ from 1 to 2=(8-1)/27=7/27.

Adapted source calculation, independently recomputed | Source: S20

Q135 • D - 0.2500

$E(X)=\int 81x^3dx=81[x^4/4]$ from 0 to 1/3=1/4.

Adapted source calculation, independently recomputed | Source: S22

Topic 19 • Bernoulli and binomial distributions

Q136 • C - Independent trials with constant success probability

Together with fixed n and binary outcomes, these define the binomial trial model.

Original question on source concepts | Source: T74-84, S28-35, S41

Q137 • B - 0 and 1

It is an indicator for failure and success.

Original question on source concepts | Source: T74-84, S28-35, S41

Q138 • A - np

It is the sum of n Bernoulli expectations p.

Original question on source concepts | Source: T74-84, S28-35, S41

Q139 • D - X≥r

The boundary r is included in at least.

Original question on source concepts | Source: T74-84, S28-35, S41

Q140 • C - 0.3125

C(5,2)×0.5^2×0.5^3=0.31250000. All fixed-count sequences are counted by the combination.

Original numerical application of source method | Source: T77-83, S33-35

Q141 • B - 0.2344

C(6,2)×0.5^2×0.5^4=0.23437500. All fixed-count sequences are counted by the combination.

Original numerical application of source method | Source: T77-83, S33-35

Q142 • A - 0.1001

C(8,0)×0.25^0×0.75^8=0.10011292. All fixed-count sequences are counted by the combination.

Original numerical application of source method | Source: T77-83, S33-35

Q143 • D - 0.2646

C(4,2)×0.3^2×0.7^2=0.26460000. All fixed-count sequences are counted by the combination.

Original numerical application of source method | Source: T77-83, S33-35

Q144 • C - 0.2684

C(10,1)×0.2^1×0.8^9=0.26843546. All fixed-count sequences are counted by the combination.

Original numerical application of source method | Source: T77-83, S33-35

Q145 • B - 0.4481

Sum the masses for 0,1,2: 0.04398047+0.15393163+0.25013890=0.44805099.

Adapted source calculation, independently recomputed | Source: T80-83

Q146 • A - 0.1094

At least five is exactly five or six: $[C(6,5)+C(6,6)]/2^6=(6+1)/64=7/64$.

Adapted source calculation, independently recomputed | Source: T78-80

Topic 20 • Poisson counts and approximation

Q147 • D - Both equal λ

The Poisson first two moments give mean and variance λ.

Original question on source concepts | Source: T85-86, S31, S35-37

Q148 • C - 0,1,2,...

Zero events is a valid Poisson outcome.

Original question on source concepts | Source: T85-86, S31, S35-37

Q149 • B - Large n and small p with λ=np

Rare independent successes motivate the limiting approximation.

Original question on source concepts | Source: T85-86, S31, S35-37

Q150 • A - Doubles it

The expected count is proportional to exposure at a constant rate.

Original question on source concepts | Source: T85-86, S31, S35-37

Q151 • D - 0.1353

exp(-2)×2^0/0!=0.13533528.

Original numerical application of source method | Source: T85-86, S35-37

Q152 • C - 0.2240

exp(-3)×3^2/2!=0.22404181.

Original numerical application of source method | Source: T85-86, S35-37

Q153 • B - 0.1378

exp(-1.6)×1.6^3/3!=0.13782802.

Original numerical application of source method | Source: T85-86, S35-37

Q154 • A - 0.1367

exp(-7.5)×7.5^6/6!=0.13671824.

Original numerical application of source method | Source: T85-86, S35-37

Q155 • D - 0.6620

At most eight includes 0 through 8: exp(-7.5) times the sum of 7.5^x/x! for those nine x values = 0.66196712. Do not replace it by P(X≥8).

Adapted source calculation, independently recomputed | Source: T85-86

Topic 21 • Geometric waiting times

Q156 • C - Trials through the first success

The convention in the source starts at 1 and includes the successful trial.

Original question on source concepts | Source: T87, S37-40, S44-45

Q157 • B - 1/p

The expected waiting time through first success is the reciprocal of success probability.

Original question on source concepts | Source: T87, S37-40, S44-45

Q158 • A - Memorylessness

Given prior failures, the remaining wait has the same distribution under constant independent trial risks.

Original question on source concepts | Source: T87, S37-40, S44-45

Q159 • D - It stays the same

Failures count equals trials count minus one, a location shift.

Original question on source concepts | Source: T87, S37-40, S44-45

Q160 • C - 0.1406

The first 2 trials fail, then one succeeds: 0.75^2×0.25=0.14062500.

Original numerical application of source method | Source: T87, S37-40

Q161 • B - 0.1024

The first 3 trials fail, then one succeeds: 0.8^3×0.2=0.10240000.

Original numerical application of source method | Source: T87, S37-40

Q162 • A - 0.2500

The first 1 trials fail, then one succeeds: 0.5^1×0.5=0.25000000.

Original numerical application of source method | Source: T87, S37-40

Q163 • D - 12.0000

Variance=(1-p)/p²=0.75/0.0625=12.

Original numerical application of source method | Source: T87, S38-40

Topic 22 • Hypergeometric sampling without replacement

Q164 • C - Hypergeometric

The finite population contains a fixed number of successes and the sample is drawn without replacement.

Original question on source concepts | Source: T87-89

Q165 • B - Population number of successes

N is population size, K is successes in it, and n is sample size.

Original question on source concepts | Source: T87-89

Q166 • A - (N-n)/(N-1)

Finite sampling without replacement reduces uncertainty in the number selected.

Original question on source concepts | Source: T87-89

Q167 • D - 0

The number of successes is then fixed at K, so there is no sampling variability.

Original question on source concepts | Source: T87-89

Q168 • C - 0.1524

P=C(4,2)C(46,8)/C(50,10)=0.15240990.

Adapted source calculation, independently recomputed | Source: T87-89

Q169 • B - 0.4696

P=C(5,1)C(15,3)/C(20,4)=0.46955624.

Original numerical application of source method | Source: T87-89

Q170 • A - 0.2182

P=C(4,2)C(8,1)/C(12,3)=0.21818182.

Original numerical application of source method | Source: T87-89

Topic 23 • Uniform or rectangular distribution

Q171 • D - 1/(b-a)

A constant height times interval length must equal one.

Original question on source concepts | Source: T89-90, S40-41

Q172 • C - (a+b)/2

Symmetry and integration place the mean at the interval midpoint.

Original question on source concepts | Source: T89-90, S40-41

Q173 • B - Equal probabilities

Their density areas are the same because the height is constant.

Original question on source concepts | Source: T89-90, S40-41

Q174 • A - $(b-a)^2/12$

This follows from the second moment minus the squared midpoint mean.

Original question on source concepts | Source: T89-90, S40-41

Q175 • D - 0.2500

Probability=interval length/support length=(4-3)/(4-0)=0.250000.

Original numerical application of source method | Source: T89-90, S40-41

Q176 • C - 0.3333

Probability=interval length/support length=(5-3)/(8-2)=0.333333.

Original numerical application of source method | Source: T89-90, S40-41

Q177 • B - 0.5000

Probability=interval length/support length=(17-12)/(20-10)=0.500000.

Original numerical application of source method | Source: T89-90, S40-41

Topic 24 • Normal distributions and standard scores

Q178 • A - Mean 0 and variance 1

The term standard fixes both location and scale.

Original question on source concepts | Source: T90-94

Q179 • D - $(X-\mu)/\sigma$

Subtract the mean and divide by the positive standard deviation.

Original question on source concepts | Source: T90-94

Q180 • C - $1-\Phi(z)$

Φ gives the left cumulative area, so subtract it from one for the right tail.

Original question on source concepts | Source: T90-94

Q181 • B - Equal to μ

The normal density is symmetric and unimodal at its mean.

Original question on source concepts | Source: T90-94

Q182 • A - 0.1075

z=(362-300)/50=1.24. Right tail=$1-\Phi(1.24)$=0.10748770.

Adapted source calculation, independently recomputed | Source: T92-94

Q183 • D - 0.1587

z=(60-50)/10=1. Right tail=$1-\Phi(1)$=0.15865525.

Adapted source calculation, independently recomputed | Source: T92-94

Q184 • C - 0.1587

z=(80-65)/15=1. Right tail=$1-\Phi(1)$=0.15865525.

Adapted source calculation, independently recomputed | Source: T92-94

Topic 25 • Gamma integrals and distribution

Q185 • B - r/λ

The supplement rate convention gives mean r divided by rate.

Original question on source concepts | Source: S42-44; T74 objective only

Q186 • A - (m-1)!

The gamma recursion extends factorials with an index shift.

Original question on source concepts | Source: S42-44; T74 objective only

Q187 • D - 1/λ

Rate and scale are reciprocals for the same gamma model.

Original question on source concepts | Source: S42-44; T74 objective only

Q188 • C - Exponential

The gamma density reduces to $\lambda \exp(-\lambda x)$ for $x>0$.

Original question on source concepts | Source: S42-44; T74 objective only

Q189 • B - 0.7500

With a rate parameter, variance=$r/\lambda^2=3/2^2=0.750000$.

Original numerical application of source method | Source: S42-44

Q190 • A - 0.1600

With a rate parameter, variance=$r/\lambda^2=4/5^2=0.160000$.

Original numerical application of source method | Source: S42-44

Q191 • D - 0.2222

With a rate parameter, variance=$r/\lambda^2=2/3^2=0.222222$.

Original numerical application of source method | Source: S42-44

Q192 • C - 0.0160

Set $t=5x$, so the result is $\Gamma(3)/5^3=2!/125=2/125$.

Adapted source calculation, independently recomputed | Source: S42

Topic 26 • Supplement bridge: descriptive revision

Q193 • B - Preserve them as source evidence while following the user's course allocation

The content is retained under the requested supplement pathway with the label discrepancy disclosed.

Original question on source concepts | Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q194 • A - Sum the individual frequencies independently

Independent summation can detect a total error without guessing at missing data.

Original question on source concepts | Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q195 • D - State one convention and apply it consistently

Quantile rules can differ; clarity and consistency make the result checkable.

Original question on source concepts | Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q196 • C - Flag them and avoid inventing missing symbols

Uncertain source material must be distinguished from explicitly added explanations.

Original question on source concepts | Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

Q197 • B - 5.2857

Sum=37 and count=7, so mean=37/7=5.285714 before rounding.

Adapted source calculation, independently recomputed | Source: S7

Q198 • A - 79.6385

Independent products sum to 51765 and frequencies to 650; mean=79.638462.

Adapted source calculation, independently recomputed | Source: S5-8

Topic 27 • Supplement bridge: sampling and standard error

Q199 • D - A statistic over repeated samples under a design

It concerns how a statistic varies across possible samples.

Original question on source concepts | Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q200 • C - σ/√n

Averaging n independent observations gives mean variance σ^2/n.

Original question on source concepts | Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q201 • B - Selection within groups need not be random

Proportionate counts alone do not give known selection probabilities.

Original question on source concepts | Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q202 • A - Random start followed by the stated fixed interval

A random start is part of the design and the interval must be respected.

Original question on source concepts | Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

Q203 • D - 2.0000

SE=σ/√n=12/√36=2.

Original numerical application of source method | Source: S48, S54

Q204 • C - 2.0000

SE=σ/√n=20/√100=2.

Original numerical application of source method | Source: S48, S54

Q205 • B - 2.0000

SE=σ/√n=8/√16=2.

Original numerical application of source method | Source: S48, S54

Topic 28 • Supplement bridge: testing and paired comparisons

Q206 • A - Before and after readings on the same units

Paired inference analyses within-unit differences rather than independent readings.

Original question on source concepts | Source: S49-61; cross-reference STA 121 topics 36-39

Q207 • D - n-1

The paired sample is the n differences, with one estimated mean.

Original question on source concepts | Source: S49-61; cross-reference STA 121 topics 36-39

Q208 • C - The prespecified alternative hypothesis

A >, < or ≠ alternative specifies the relevant tail or tails.

Original question on source concepts | Source: S49-61; cross-reference STA 121 topics 36-39

Q209 • B - The test did not provide sufficient evidence against H0

A nonsignificant result is a statement about evidence under the test, not proof of the null.

Original question on source concepts | Source: S49-61; cross-reference STA 121 topics 36-39

Q210 • A - 4.2426

Mean difference=3, sample variance=2.5, SE=√(2.5/5)=0.707107; t=4.242641, df=4.

Original numerical application of source method | Source: S55

Topic 29 • Analysis of variance and post-hoc comparisons

Q211 • D - At least one population mean differs

An omnibus test detects some departure from equality without identifying every pair.

Original question on source concepts | Source: S60, S62-63, S65

Q212 • C - Between-group mean square divided by error mean square

The mean-square ratio measures between-group variation relative to within-group variation.

Original question on source concepts | Source: S60, S62-63, S65

Q213 • B - N-k

Within each group a mean is estimated, using k degrees of freedom overall.

Original question on source concepts | Source: S60, S62-63, S65

Q214 • A - Identify particular group differences while addressing multiple comparisons

Procedures such as Tukey investigate pairs with a multiplicity-aware method.

Original question on source concepts | Source: S60, S62-63, S65

Q215 • D - 18

df error=N-k=22-4=18.

Adapted source calculation, independently recomputed | Source: S62

Q216 • C - 36.0000

Group means=3,6,9 and grand mean=6. SSB=2[(3-6)²+(6-6)²+(9-6)²]=36.

Original numerical application of source method | Source: S62, S65

Q217 • B - 9.0000

MSB=36/2=18; MSE=6/3=2; F=18/2=9.

Original numerical application of source method | Source: S62, S65

Topic 30 • Chi-square goodness of fit and independence

Q218 • A - Observed category counts versus specified probabilities

Expected counts under the null provide a model for the categorical frequencies.

Original question on source concepts | Source: S55, S63-65

Q219 • D - Row total times column total divided by grand total

This is the count implied by the product of marginal proportions.

Original question on source concepts | Source: S55, S63-65

Q220 • C - (r-1)(c-1)

Fixed row and column totals constrain the independent cell counts.

Original question on source concepts | Source: S55, S63-65

Q221 • B - Evidence of association under the model

The test concerns association of categorical variables, not a causal mechanism.

Original question on source concepts | Source: S55, S63-65

Q222 • A - 0.4000

Total=80, expected=20 each. Sum=4/20+4/20=0.4.

Original numerical application of source method | Source: S55, S63-64

Q223 • D - 6

df=(3-1)(4-1)=2×3=6.

Original numerical application of source method | Source: S55, S63-64

Topic 31 • Experimental design and correlation concepts

Q224 • C - Randomisation

Chance assignment reduces systematic treatment allocation differences.

Original question on source concepts | Source: S65; S60, S62 contextual mentions

Q225 • B - Replication

Independent treatment replication permits estimation of experimental variability.

Original question on source concepts | Source: S65; S60, S62 contextual mentions

Q226 • A - Local control

Blocking manages nuisance variation by grouping similar units in the design.

Original question on source concepts | Source: S65; S60, S62 contextual mentions

Q227 • D - Association, with possible confounding

Association can result from common causes or reverse direction.

Original question on source concepts | Source: S65; S60, S62 contextual mentions

STA 121 • worked answers

Topic 32 • Sampling foundations and simple random sampling

Q228 • C - The operational list or structure used to select units

It links the target population to an actual selection procedure.

Original question on source concepts | Source: T95-98

Q229 • B - Every possible sample of that size is equally likely

Equal probabilities across all fixed-size subsets define simple random sampling without replacement.

Original question on source concepts | Source: T95-98

Q230 • A - Reject the repeated selection and continue by the rule

A unit cannot appear twice in a without-replacement sample.

Original question on source concepts | Source: T95-98

Q231 • D - They do not automatically remove selection bias

More observations can reduce variability while preserving a systematic selection problem.

Original question on source concepts | Source: T95-98

Q232 • C - 120

The number of subsets is C(10,3)=10!/(3!7!)=120.

Original numerical application of source method | Source: T96-98

Q233 • B - 0.2500

Every unit has inclusion probability n/N=5/20=0.25.

Original numerical application of source method | Source: T96-98

Topic 33 • Stratified and cluster sampling

Q234 • A - nNh/N

Each stratum receives the same sampling fraction n/N.

Original question on source concepts | Source: T98-100

Q235 • D - All units in the chosen clusters

Selecting whole clusters distinguishes this one-stage design from within-cluster subsampling.

Original question on source concepts | Source: T98-100

Q236 • C - Mutually exclusive and collectively cover the population

Every unit should belong to exactly one stratum for this partition design.

Original question on source concepts | Source: T98-100

Q237 • B - Units within clusters can be similar

Within-cluster correlation can reduce the information from a fixed number of sampled units.

Original question on source concepts | Source: T98-100

Q238 • A - 14

nh=nNh/N=50×140/500=14.

Original numerical application of source method | Source: T98-103

Q239 • D - 20

nh=nNh/N=60×200/600=20.

Original numerical application of source method | Source: T98-103

Q240 • C - 70

nh=nNh/N=200×420/1200=70.

Original numerical application of source method | Source: T98-103

Topic 34 • Systematic and multistage sampling

Q241 • B - N/n

A step of N/n spaces n selections over the ordered frame.

Original question on source concepts | Source: T101-103

Q242 • A - Periodicity aligned with the interval

A repeating pattern can make selected positions unrepresentative.

Original question on source concepts | Source: T101-103

Q243 • D - Multistage sampling

Units are selected at successive hierarchical stages.

Original question on source concepts | Source: T101-103

Q244 • C - Further size and selection conditions make them equal

Student inclusion depends on school selection and conditional selection inside schools.

Original question on source concepts | Source: T101-103

Q245 • B - 35

k=N/n=10. Fourth position=r+3k=5+3×10=35.

Original numerical application of source method | Source: T101-102

Q246 • A - 20

k=N/n=6. Fourth position=r+3k=2+3×6=20.

Original numerical application of source method | Source: T101-102

Q247 • D - 37

k=N/n=10. Fourth position=r+3k=7+3×10=37.

Original numerical application of source method | Source: T101-102

Topic 35 • Quota and purposive sampling

Q248 • C - Quota sampling

Quota targets do not make selection within groups random.

Original question on source concepts | Source: T102-103

Q249 • B - Purposive sampling

The choice is based on relevance and judgement rather than known random selection probabilities.

Original question on source concepts | Source: T102-103

Q250 • A - Only that the chosen quotas were filled

Balance on selected categories does not ensure balance on all characteristics.

Original question on source concepts | Source: T102-103

Q251 • D - Population estimation with design-based uncertainty

Known selection probabilities support probability-based population estimation.

Original question on source concepts | Source: T102-103

Q252 • C - 21

Quota=0.7×30=21. This is a count target, not evidence of random selection.

Original numerical application of source method | Source: T102-103

Q253 • B - 20

Quota=0.4×50=20. This is a count target, not evidence of random selection.

Original numerical application of source method | Source: T102-103

Topic 36 • Hypotheses, errors and decisions

Q254 • A - A true H0 is rejected

Type I concerns a mistaken rejection under a true null.

Original question on source concepts | Source: T104-108

Q255 • D - 1-β

Power is the probability of rejecting H0 at a specified alternative.

Original question on source concepts | Source: T104-108

Q256 • C - H0 and the test model

It is a null-model tail probability, not the probability that the null is true.

Original question on source concepts | Source: T104-108

Q257 • B - Absolute z exceeds 1.96

A total α=0.05 is split between two tails.

Original question on source concepts | Source: T104-108

Q258 • A - 0.8000

Power=1-β=1-0.2=0.8.

Original numerical application of source method | Source: T108

Q259 • D - 0.6500

Power=1-β=1-0.35=0.65.

Original numerical application of source method | Source: T108

Q260 • C - 0.9000

Power=1-β=1-0.1=0.9.

Original numerical application of source method | Source: T108

Topic 37 • One-sample tests for a population mean

Q261 • B - t=(mean-μ0)/(s/√n)

The sample SD estimates σ, giving a t distribution with n-1 degrees of freedom.

Original question on source concepts | Source: T108-113

Q262 • A - σ/√n

The mean has variance σ^2/n for independent observations.

Original question on source concepts | Source: T108-113

Q263 • D - A sufficiently negative t

The rejection region follows the lower-direction alternative.

Original question on source concepts | Source: T108-113

Q264 • C - It is a classroom heuristic, not the definition of t versus z

Distributional assumptions and whether σ is known matter in choosing the procedure.

Original question on source concepts | Source: T108-113

Q265 • B - 2.1082

SE=0.36/√10=0.113842; t=(2.24-2)/SE=2.108185, df=9.

Adapted source calculation, independently recomputed | Source: T109-112

Q266 • A - 2.0000

SE=10/√25=2.000000; t=(54-50)/SE=2.000000, df=24.

Original numerical application of source method | Source: T109-112

Q267 • D - -2.0000

SE=4/√16=1.000000; t=(18-20)/SE=-2.000000, df=15.

Original numerical application of source method | Source: T109-112

Topic 38 • Tests comparing two population means

Q268 • C - They are equal

The pooled variance estimate combines two estimates of a common variance.

Original question on source concepts | Source: T113-116

Q269 • B - n1+n2-2

Two sample means are estimated across the two samples.

Original question on source concepts | Source: T113-116

Q270 • A - Matched before-and-after data

Pairing creates a dependence structure addressed by analysing differences.

Original question on source concepts | Source: T113-116

Q271 • D - Square root of $\sigma1^2/n1+\sigma2^2/n2$

Independent sample-mean variances add, and SE is their square root.

Original question on source concepts | Source: T113-116

Q272 • C - 2.8284

SE=√(25/25+25/25)=√2; z=4/√2=2.828427.

Original numerical application of source method | Source: T113-114

Q273 • B - 12

df=n1+n2-2=8+6-2=12.

Original numerical application of source method | Source: T114

Topic 39 • Tests for one and two proportions

Q274 • A - p0(1-p0)

The null model determines the standard error for the test.

Original question on source concepts | Source: T116-120

Q275 • D - (x1+x2)/(n1+n2)

Pool successes and observations, rather than averaging sample proportions without weights.

Original question on source concepts | Source: T116-120

Q276 • C - Unpooled estimated SE

An interval estimates both population proportions rather than imposing their equality.

Original question on source concepts | Source: T116-120

Q277 • B - 5

Three percent to eight percent differs by five percentage points, although the proportion difference is 0.05.

Original question on source concepts | Source: T116-120

Q278 • A - 1.0206

phat=0.65; null SE=√(0.60×0.40/100)=0.0489898. z=1.020621.

Adapted source calculation, independently recomputed | Source: T118-119

Q279 • D - 0.0591

Pool counts and denominators: (30+100)/(1000+1200)=130/2200=0.059091.

Adapted source calculation, independently recomputed | Source: T117, T120

Q280 • C - -3.5355

phat=0.75; null SE=√(0.8×0.2/800)=0.01414214. z=(0.75-0.8)/SE=-3.535534.

Adapted source calculation, independently recomputed | Source: T120 exercise 5

Topic 40 • Point estimation and properties of estimators

Q281 • B - A rule applied to random sample data

Before sampling, the rule's value varies with possible samples.

Original question on source concepts | Source: T121-124

Q282 • A - The estimator's expectation equals the parameter

It is a repeated-sampling property of the estimator.

Original question on source concepts | Source: T121-124

Q283 • D - Convergence to the parameter as sample size grows

A consistent estimator approaches the target in probability as n increases.

Original question on source concepts | Source: T121-124

Q284 • C - Greater efficiency in that comparison

Efficiency compares sampling variability under the stated conditions.

Original question on source concepts | Source: T121-124

Q285 • B - 2.2500

The ratio is 9/4=2.25; U has the smaller variance in this comparison.

Original numerical application of source method | Source: T123

Topic 41 • Confidence intervals for means and proportions

Q286 • A - Critical multiplier times SE

The interval is estimate plus or minus this margin.

Original question on source concepts | Source: T125-128, T132-133

Q287 • D - Long-run coverage of the interval procedure

Across repeated samples, about 95% of intervals from that procedure cover the fixed parameter.

Original question on source concepts | Source: T125-128, T132-133

Q288 • C - Increases it

A larger critical multiplier is needed for greater coverage.

Original question on source concepts | Source: T125-128, T132-133

Q289 • B - A t critical value with n-1 df

The t multiplier accounts for estimating the population SD.

Original question on source concepts | Source: T125-128, T132-133

Q290 • A - 48.0400

SE=10/√100=1; margin=1.96×SE=1.96; lower endpoint=50-1.96=48.04.

Original numerical application of source method | Source: T125-127

Q291 • D - 28.0400

SE=8/√64=1; margin=1.96×SE=1.96; lower endpoint=30-1.96=28.04.

Original numerical application of source method | Source: T125-127

Q292 • C - 18.0400

SE=6/√36=1; margin=1.96×SE=1.96; lower endpoint=20-1.96=18.04.

Original numerical application of source method | Source: T125-127

Q293 • B - -0.0345

Difference=-0.05333333. Unpooled SE=√[0.03(0.97)/1000+(1/12)(11/12)/1200]=0.00963106. Upper endpoint=difference+1.96SE=-0.03445645.

Adapted source calculation, independently recomputed | Source: T133 exercise 7

Topic 42 • Sample size and precision planning

Q294 • A - Up to the next integer

Rounding down can violate the requested margin of error.

Original question on source concepts | Source: T128-133

Q295 • D - 0.5

p(1-p) is maximised at 0.5, giving the largest required n for fixed precision.

Original question on source concepts | Source: T128-133

Q296 • C - Multiply by 4

Margin scales as 1/√n, so its inverse square determines n.

Original question on source concepts | Source: T128-133

Q297 • B - Selection bias

A larger but systematically selected sample can preserve the same bias.

Original question on source concepts | Source: T128-133

Q298 • A - 186

n=(1.96×6.95/1)²=185.558884. Round up to 186.

Adapted source calculation, independently recomputed | Source: T129-131

Q299 • D - 97

n=(1.96×10/2)²=96.040000. Round up to 97.

Original numerical application of source method | Source: T129-131

Q300 • C - 97

n=(1.96×5/1)²=96.040000. Round up to 97.

Original numerical application of source method | Source: T129-131

Fill-in-the-gap answers

STA 111 • Topic 01 • Statistical thinking and inference

F001 • parameter

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T1-4

F002 • inductive

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T1-4

STA 111 • Topic 02 • Variables and the meaning of data

F003 • identifier

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T4-6, T9

F004 • discrete

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T4-6, T9

STA 111 • Topic 03 • Data sources, collection and quality

F005 • primary

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T5-8

F006 • experiment

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T5-8

STA 111 • Topic 04 • Official statistics and responsible use

F007 • denominator

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T8-9

F008 • CBN

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T8-9

STA 111 • Topic 05 • Frequency tables and statistical graphs

F009 • total

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T10-15, T43-47

F010 • ogive

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T10-15, T43-47

F011 • 26

3+4+8+11=26. Source: T10-15, T43-47

STA 111 • Topic 06 • Stem-and-leaf displays and box plots

F012 • interquartile

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T14-17, T44

F013 • key

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T14-17, T44

STA 111 • Topic 07 • Arithmetic, harmonic and geometric means

F014 • harmonic

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T18-21

F015 • geometric

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T18-21

F016 • 83

0+3+4+15+8+10+6+21+16=83. Source: T18-21

STA 111 • Topic 08 • Median, mode and robust location

F017 • mode

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T21-25

F018 • median

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T21-25

F019 • 7

Average the third and fourth values: (6+8)/2=7. Source: T21-25

STA 111 • Topic 09 • Quartiles, deciles and percentiles

F020 • median

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T25-28

F021 • 75th

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T25-28

F022 • 4

(12-4)/2=4. Source: T25-28

STA 111 • Topic 10 • Dispersion, variance and relative variation

F023 • standard deviation

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T28-34

F024 • n-1

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T28-34

F025 • 8

Mean=7; SS=40; divide by 5. Source: T28-34

STA 111 • Topic 11 • Skewness, kurtosis and summary output

F026 • positive

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T34-42

F027 • 3

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T34-42

STA 112 • Topic 12 • Random variables and probability models

F028 • values

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T48-51, S12, S15

F029 • 0

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T48-51, S12, S15

STA 112 • Topic 13 • Expectation, variance and transformations

F030 • square

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T52-55, S15-17, S21-26

F031 • b^2

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T52-55, S15-17, S21-26

F032 • 1

$10-3^2=1$. Source: T52-55, S15-17, S21-26

STA 112 • Topic 14 • Events, axioms and probability laws

F033 • intersection

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T56-61, S12, S27

F034 • exclusive

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T56-61, S12, S27

F035 • 0.7

0.4+0.5-0.2=0.7. Source: T56-61, S12, S27

STA 112 • Topic 15 • Conditional probability, tables and replacement

F036 • B

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T62-72

F037 • dependent

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T62-72

STA 112 • Topic 16 • Combinatorial analysis

F038 • combination

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S13-14, S66

F039 • 1

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S13-14, S66

F040 • 252

C(10,5)=252. Source: S13-14, S66

STA 112 • Topic 17 • Discrete cumulative distribution functions

F041 • continuous

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S17-18; T49-50, T74-77

F042 • jump

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S17-18; T49-50, T74-77

STA 112 • Topic 18 • Continuous densities and cumulative probabilities

F043 • 1

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T51, T54-55, S18-20, S22-26

F044 • density

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T51, T54-55, S18-20, S22-26

STA 112 • Topic 19 • Bernoulli and binomial distributions

F045 • Bernoulli

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T74-84, S28-35, S41

F046 • binomial

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T74-84, S28-35, S41

F047 • 2

np=8×0.25=2. Source: T74-84, S28-35, S41

STA 112 • Topic 20 • Poisson counts and approximation

F048 • λ

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T85-86, S31, S35-37

F049 • np

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T85-86, S31, S35-37

F050 • 3

Poisson variance equals its parameter λ. Source: T85-86, S31, S35-37

STA 112 • Topic 21 • Geometric waiting times

F051 • 1

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T87, S37-40, S44-45

F052 • memoryless

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T87, S37-40, S44-45

F053 • 5

1/p=1/0.2=5. Source: T87, S37-40, S44-45

STA 112 • Topic 22 • Hypergeometric sampling without replacement

F054 • hypergeometric

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T87-89

F055 • 0

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T87-89

STA 112 • Topic 23 • Uniform or rectangular distribution

F056 • 1/(b-a)

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T89-90, S40-41

F057 • (a+b)/2

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T89-90, S40-41

STA 112 • Topic 24 • Normal distributions and standard scores

F058 • 1

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T90-94

F059 • standardisation

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T90-94

STA 112 • Topic 25 • Gamma integrals and distribution

F060 • reciprocals

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S42-44; T74 objective only

F061 • (m-1)!

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S42-44; T74 objective only

F062 • 1.5

r/λ=3/2=1.5. Source: S42-44; T74 objective only

STA 112 • Topic 26 • Supplement bridge: descriptive revision

F063 • 111

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

F064 • summation

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S1-11, S69-70; cross-reference STA 111 topics 1-11

STA 112 • Topic 27 • Supplement bridge: sampling and standard error

F065 • error

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

F066 • N

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S46-48, S53-54, S61, S64; cross-reference STA 121 topics 32-35

STA 112 • Topic 28 • Supplement bridge: testing and paired comparisons

F067 • differences

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S49-61; cross-reference STA 121 topics 36-39

F068 • ≠

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S49-61; cross-reference STA 121 topics 36-39

STA 112 • Topic 29 • Analysis of variance and post-hoc comparisons

F069 • error

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S60, S62-63, S65

F070 • k

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S60, S62-63, S65

F071 • 18

N-k=22-4=18. Source: S60, S62-63, S65

STA 112 • Topic 30 • Chi-square goodness of fit and independence

F072 • E

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S55, S63-65

F073 • grand

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S55, S63-65

STA 112 • Topic 31 • Experimental design and correlation concepts

F074 • randomisation

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S65; S60, S62 contextual mentions

F075 • control

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: S65; S60, S62 contextual mentions

STA 121 • Topic 32 • Sampling foundations and simple random sampling

F076 • frame

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T95-98

F077 • probability

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T95-98

STA 121 • Topic 33 • Stratified and cluster sampling

F078 • every

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T98-100

F079 • clusters

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T98-100

STA 121 • Topic 34 • Systematic and multistage sampling

F080 • n

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T101-103

F081 • multistage

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T101-103

STA 121 • Topic 35 • Quota and purposive sampling

F082 • quota

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T102-103

F083 • purposive

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T102-103

STA 121 • Topic 36 • Hypotheses, errors and decisions

F084 • I

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T104-108

F085 • β

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T104-108

F086 • 0.8

Power=1-β=0.8. Source: T104-108

STA 121 • Topic 37 • One-sample tests for a population mean

F087 • n-1

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T108-113

F088 • negative

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T108-113

STA 121 • Topic 38 • Tests comparing two population means

F089 • variances

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T113-116

F090 • 2

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T113-116

STA 121 • Topic 39 • Tests for one and two proportions

F091 • null

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T116-120

F092 • samples

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T116-120

STA 121 • Topic 40 • Point estimation and properties of estimators

F093 • parameter

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T121-124

F094 • consistency

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T121-124

STA 121 • Topic 41 • Confidence intervals for means and proportions

F095 • 2

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T125-128, T132-133

F096 • coverage

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T125-128, T132-133

F097 • 4

Width is twice the margin. Source: T125-128, T132-133

STA 121 • Topic 42 • Sample size and precision planning

F098 • up

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T128-133

F099 • 0.5

This follows the definition or rule developed in this topic; equivalent correctly stated terminology is accepted. Source: T128-133

F100 • 385

$1.96^2 \times 0.25/0.05^2 = 384.16$, rounded up to 385. Source: T128-133

Question-bank source policy

All items were authored for this study edition. A reference indicates the source concept or exercise that supports the item; it does not imply that original distractors or new numerical inputs appear verbatim in the scans. Adapted source calculations are identified in the solutions. The bank is written as introductory university practice for Adeleke University students; no official examination pattern or past-paper authenticity is claimed.

Excluded or corrected inputs

T11-19 • The frequency table has ∑fx=83, not 166. Its mean is 83/24. The grouped table is a separate dataset with mean 2532.5/45.

T25-28 • Use fractional rank remainder in ungrouped interpolation and pN for grouped percentiles. T26 says grams while T27 reports kilograms; preserve numeric scale without resolving units by assumption.

T30-37 • Some variance/SD labels are interchanged. The crate table has SS=830; population variance=27.6667. Its grouped mode is 13.6667 and Pearson first skewness about 0.0634.

T55 exercise 3 • Printed probabilities 1/2,1/6,2/3 sum to 4/3. Excluded as an ordinary moment exercise; not silently renormalised.

T61 • Simplified addition applies to mutually exclusive events, not to arbitrary independent events.

T70-72 • The resistor box composition is absent. The burglary example lacks a joint model. Do not invent the missing counts or independence.

T80-83 • For Bin(14,0.2), checked values are P(X≤2)=0.448051, P(X≥5)=0.129840 and P(2≤X≤4)=0.672248.

T88-89 • The first hypergeometric question states four defectives but substitutes five. The following question states N=150,K=20,n=50,x=5 but substitutes other values. Use the question-stated values explicitly.

T89-90 • The conference prose mentions six hours, then explicitly models [0,4]. Only the stated [0,4] model is used for that worked calculation.

T92 • The normal example asks above 362 but changes labels to 365. The worked answer in this pack explicitly solves 362.

T110 • For the given summaries, t=2.1082 with df=9 does not exceed two-sided 5% critical 2.262. Some raw values are cropped; the summaries are the chosen inputs.

T111-116 • Retain the sign of a t statistic. The bank-data t is -1.9044. The faculty pooled variance is 45.2917; printed SE and pooled arithmetic contain errors.

T113 • The prose states 400 men while a worked line says N=20. A clearly labelled reconstruction may use 400; no ambiguous item relies on the stray 20.

T117-119 • Pooled denominator is n1+n2. The one-proportion test uses p0 in its null SE; the gambler statistic is 1.0206.

S5-8 • Frequencies sum to 650. Recomputed ∑fx=51765 and mean=79.6385. Some row products and the S5 final product total disagree; the S6 grouped median and mode are checkable.

S13 • 18P6=13366080; the written numeric result is inconsistent with the correct formula.

S17-20 • Overlapping endpoint inequalities are rebuilt as right-continuous CDFs. Use density $x^2/9$ for the S20 integral; $x^3/27$ is its CDF.

S35 • For five fair tosses P(X≤4)=31/32, not the single X=5 mass. At least and at most are interpreted literally.

S62-65 • Faculty labels and overwritten values are uncertain. The chi-square worked O/E totals are inconsistent. Teach their methods using fully specified original illustrations; no reconstructed original numeric test is claimed.

S67-70 • S67 logic gates and S68 implicit differentiation are excluded from statistics. S69-70 have clipped formula fragments and stray algebra; use only verified statistical identities.

Consult the source map and complete correction register in the companion lesson PDF. All page citations refer to source PDF page positions, not pages of this question bank.